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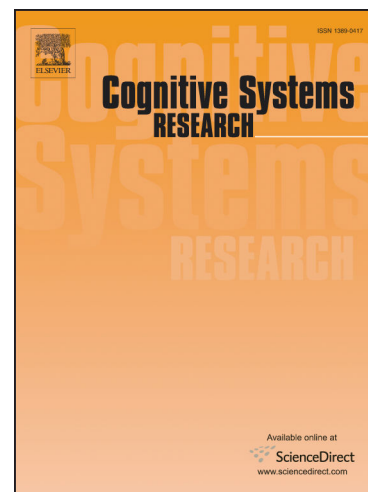
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Supporting Risk Management through Cyberspace: An Adaptive Network Model Simulating AI Coach Effects by Inducing Adherence to Guidelines in Neonatal Medical Protocols

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Abstract. In this article, it is shown how second-order adaptive agent-based network models can be used to support a medical team in healthcare institutions to adhere to specific Neonatal Hypoglycemia and Neonatal Hyperbilirubinemia treatment guidelines through the integration of an Artificial Intelligence (AI) Virtual Coach. The proposed AI Coach is designed to provide timely interventions and correct deviations when lapses in the health care practitioner's internal mental model occur. Through simulating three different scenarios, the internal dynamics of these mental models, adaptive changes of these mental models (learning and forgetting), and the interaction between health care practitioners and the world is shown when: (1) There is perfect adherence to guidelines, (2) There is imperfect adherence to guidelines and (3) There is both perfect and imperfect adherence to guidelines alongside interventions of the AI Coach in the latter case.

1 Introduction

Healthcare services provided at any care centre, are based upon the laid-out guidelines of medical protocols. These operating procedures play a critical role in ensuring that healthcare standards are justly maintained and that patients receive the most optimal care. In an ideal world, such guidelines would be followed perfectly without any mistakes committed. Often however, practically due to the nature of human error, deviations from established protocols by medical professionals can lead to suboptimal patient care, medical errors and even compromise the safety of clinical subjects. The complexity of human behavior and cognitive processes makes monitoring, predicting, and enforcing adherence to these protocols challenging. Considering this, the introduction of Artificial Intelligence (AI) offers promising capabilities for mitigating this issue. The research embodied in this article explores the potential of a virtual AI Coach to foster adherence to medical protocols, by providing timely

intervention in instances of deviation by medical professionals. By drawing upon the modeling approach of self-modeling network modeling (Treur, 2020), this paper aims to integrate these principles to construct mental models in a temporal-causal network of a medical health care practitioner (HCP) and an AI Coach (AIC), to simulate and explore the effect of the AI on the intricate interactions that dictate adherence to various medical protocols in the domain of neonatal care.

2 Background Literature

This section delivers an overview of the background literature and concepts, serving as a foundational basis for the experimental simulations that the paper entails.

2.1 Neonatal Hypoglycemia

Glucose is a type of sugar that serves as a vital energy source for the metabolic processes in the brain. Hypoglycemia is the condition of having an abnormally low blood sugar (glucose) concentration in the body (Hawdon, 2019). Depending on its intensity and persistence, Hypoglycemia may lead to unfavourable neurological effects which can have potentially long-term implications if not treated on time. Clinically, Neonatal Hypoglycemia is defined when a newborn neonate has a blood sugar level of less than 2.5 mmol/L (45 mg/dL) and a glucose cut-off threshold of above 2.6 mmol/L (47 mg/dL) is considered to be clinically out of risk if glucose levels remain stable for three consecutive monitor readings (Rao, 2022) (Rasmussen et al., 2017).

The cornerstone of Hypoglycemia prevention is foremostly the identification of at-risk neonates. There are significant challenges associated with clinical detection of this condition when a neonate is born as quite often, the infants show no visible signs of Hypoglycemia and are otherwise healthy neonates (asymptomatic) (Rasmussen et al., 2017). The module protocol guideline followed while creating the network-oriented self-model concerns preventing Neonatal Hypoglycemia in this group of otherwise healthy neonates who are at risk for developing Hypoglycemia within the first 24 hours after birth without showing any external symptoms. These neonates adhere to a different set of guidelines for preventing and treating Hypoglycemia when compared to neonates who show visible symptoms of Hypoglycemia such as sweating, feeding difficulties, poor suck, weak or high-pitched cry, tremors, hypothermia, and seizures (Abramowski and Hamdan, 2021). Asymptomatic otherwise healthy neonates are likely to develop neonatal hypoglycemia if they are born having major risk factors, including prematurity (gestational age <37 weeks), small for gestational age (SGA, birth weight < 10th percentile of neonates), large for gestational age (LGA, birth weight > 90th percentile of neonates), and infants born to mothers with diabetes (type 1, type 2) who received insulin or oral blood sugar lowering medication during pregnancy (NVK, 2020).

The susceptibility of these neonates to neonatal hypoglycemia arises from a combination of factors. Firstly, there is a limited capacity for endogenous glucose production, which is further compounded by increased glucose consumption and inadequate food intake during the initial days following birth. This precarious metabolic situation, disrupts the normal glucose metabolism adaptation in neonates and hampers their access to alternative fuel sources, thus elevating the risk of potential brain damage

(NVK, 2020). Timely feeding, preferably breastfeeding, is often helpful in preventing hypoglycemia in high-risk otherwise healthy neonates (NVK, 2020). Once diagnosed with Neonatal Hypoglycemia, depending on the severity of glucose concentration levels in the blood, the preventive measures taken would be from a heavy or normal intravenous (IV) glucose dose to raising nutritional intake of carbohydrates for less severe glucose levels (NVK, 2020). It is important to note that various types of intravenous (IV) therapies exist. To treat the most severe hypoglycemia cases, a 'bolus' or a push IV dose is given. Here a heavy glucose dose is injected through a syringe in a short span of time. There also exist continuous and drip IV therapies, which either continuously feed glucose intravenously over a period of time or keep giving small doses after certain time intervals respectively (NVK,2020). In the treatment of Hypoglycemia, a high dose and a normal dose of continuous IV therapy are used. The treatment of neonatal hypoglycemia presents several vulnerabilities that can become a cause for healthcare practitioners to deviate from following parts of the medical protocol inaccurately. One of the biggest weaknesses concerns late or missed diagnosis of risk factors for hypoglycemia leading to missed or delayed diagnosis especially for neonates who are asymptomatic, thus exacerbating long-term neurodevelopmental outcomes (McKinlay, Alsweiler, and Harding, 2015).

Thus, it is vital to screen every neonate to see if the neonate has any risk factors and observe that these warrant further precaution, enough to start following the medical protocol. The issue of monitoring is also critical in managing neonatal hypoglycemia. Insufficient monitoring and improper blood sampling can lead to unnoticed episodes of hypoglycemia, which may result in sub-optimal treatment approaches (Thornton et al., 2015). Moreover, the management protocols also demonstrate vulnerabilities in treatment. Deviations from treatment guidelines or delays in implementing necessary interventions, such as a bolus glucose dose before intravenous glucose therapy in the case of severe hypoglycemia, can prolong the onset of the condition in the neonate, causing potential harm to the infant (Adamkin, 2017). Furthermore, the importance of an appropriate follow-up after glucose levels stabilise outside the risk zone is essential but is often not given the attention it deserves. Without proper post-treatment surveillance, clinicians may fail to identify persistent symptoms or observe a potential relapse into hypoglycemia, thus failing to provide optimal care (Narvey and Mark, 2019). These vulnerabilities in neonatal hypoglycemia management emphasize the need for rigorous adherence to guidelines, accurate and frequent monitoring, appropriate treatment, and comprehensive follow-up care.

2.2 Neonatal Hyperbilirubinemia

Hyperbilirubinemia, colloquially known as Jaundice, is a medical condition characterized by a visible yellowish discoloration of the skin, sclera, and mucous membranes due to abnormally high concentrations of serum bilirubin levels combined with a simultaneously low bilirubin conjugation capacity in the liver (Ullah et al., 2016). Bilirubin is a yellow pigmented molecule that is a product resulting from the catabolism of haem-containing compounds such as haemoglobin in the blood. It is a known toxic metabolite and should normally be processed and conjugated by the liver with glucuronic acid into bilirubin glucuronides as part of bile and excreted out of the body. However, this process is hindered in individuals with liver deficiencies as mentioned

above. High bilirubin levels cause blood plasma and thus subsequently the skin to develop a yellow tint, being one of the distinct symptoms of Jaundice.

It is important to note that bilirubin exists in the blood in 2 types, namely, indirect bilirubin also known as unconjugated bilirubin which is not water soluble and thus not measurable, and direct bilirubin also known as conjugated bilirubin which is water soluble after being conjugated as indirect bilirubin in the liver. Total bilirubin also known as Serum Bilirubin (SBR) is the total bilirubin in the blood plasma of the body and encompasses the sum of the 2 types described above (Laxmi et al., 2018).

Neonatal Hyperbilirubinemia is a cause of major concern as 9% to 11% of neonates develop upon birth with 2% experiencing severe Jaundice (Ullah et al., 2016). Severe Hyperbilirubinemia can lead to potentially lethal complications. A notable example of this is Kernicterus, a neurological syndrome resulting from bilirubin toxicity that can inflict permanent damage to the basal ganglia, hippocampus, and cranial nerve nuclei (Johnson and Bhutani, 2011). Clinically, Jaundice is diagnosed in neonates when serum bilirubin level is observed to be above 5 mg/dL, whereas severe Jaundice is recognised when serum bilirubin levels reach 20 mg/dL (Slusher et al., 2017). Premature infants are particularly vulnerable to Hyperbilirubinemia, owing to their immature hepatic function, increased bilirubin production, and enhanced enterohepatic circulation (AAP, 2004). There are several observed symptoms which point to the developing concentration levels of bilirubin in blood plasma. These include, foremostly visually observable yellow tinge of the skin and eyes, followed by dark urine, pale stools and maternal blood group incompatibility (Gillot, 2017). Maternal blood group incompatibility in particular can give rise to various conditions such as rhesus disease, and hemolytic disease which are linked to the occurrence of jaundice in neonates (Gillot, 2017).

Current clinical protocol necessitates monitoring of serum bilirubin levels and tracking the infant's jaundice visually (AAP, 2004). Depending on the severity of the concentration of total serum bilirubin (TSB), different treatment strategies are used which include various levels of phototherapy and, in severe cases, exchange transfusion. Phototherapy uses light waves to allow for the absorption of light through the skin which converts unconjugated bilirubin into bilirubin photoproducts that are excreted in the stool and urine (Muchowski, 2014). Single phototherapy refers to phototherapy with a single light source, however with higher bilirubin levels, double phototherapy is used to cover a larger surface area of the baby's body which will further help bring down the bilirubin levels in the body. Intensive phototherapy implies irradiance in the blue-green spectrum with wavelengths of 460nm to 490nm and is the most effective in converting unconjugated bilirubin (Muchowski, 2014).

Despite its efficacy, phototherapy is not without drawbacks. Among these are potential side effects such as dehydration, due to increased fluid loss through the skin, and skin rashes. In addition, a less common yet critical condition known as bronze baby syndrome may develop in infants with liver disease undergoing phototherapy. This condition is characterized by a greyish-brown discoloration of the skin. Phototherapy also necessitates a physical separation between the infant and the mother, causing an

emotional and psychological toll on the parents particularly in a period where parent-infant bonding is critical (Muchowski, 2014).

In severe cases of hyperbilirubinemia, exchange transfusion might be required. Exchange transfusion is a complex medical procedure, during which the baby's blood is gradually removed and replaced with fresh donor blood or plasma (Maisels, 2009). This procedure aims to rapidly decrease bilirubin levels in the blood and prevent or treat acute bilirubin encephalopathy, a potentially life-threatening condition. Specifically, infants presenting with TSB levels of 25 mg/dL or higher, along with symptoms of jaundice and signs of acute bilirubin encephalopathy, are considered for this intervention (Muchowski, 2014).

However, exchange transfusion is not a risk-free procedure. Mortality within the first six hours post-procedure in full-term infants without hemolysis is between three and four per 1,000 infants (Muchowski, 2014). Moreover, approximately 5% of infants undergoing exchange transfusion may experience serious complications, including blood-related problems like changes in blood clotting factors, cardiorespiratory instability, metabolic abnormalities such as electrolyte imbalances and complications from central line placement (Muchowski, 2014). Thus, the judicious application of these strategies is crucial as unwarranted treatment could expose the neonate to unnecessary risks including Hemolysis and Sepsis.

The guidelines laid out for preventing neonatal hyperbilirubinemia present a variety of vulnerabilities that may compromise the effectiveness of its treatment. One such vulnerability is the failure to observe jaundice symptoms visually. In particular, the possibility of not realising that the skin of a neonate has tinged into a light shade of yellow (a physical sign of hyperbilirubinemia) is likely, and could lead to an insufficient or late response, potentially causing serious complications (Keren et al., 2009). Ceasing phototherapy prematurely is another vulnerability in the treatment of neonatal hyperbilirubinemia as abrupt or premature discontinuation of phototherapy without proper post-treatment supervision could lead to a rebound increase in bilirubin levels, thereby precipitating a need for re-treatment (Paul and Maisels, 2017). It is also equally important to dynamically reduce and increase the treatment intensity by routinely following and checking the serum bilirubin levels (Yetman et al., 1998). It is important not to continue intensive treatment such as exchange transfusion and intensive phototherapy for prolonged times, while automatically reducing the treatment threshold from intensive to a lighter single or double phototherapy alongside frequently checking total serum bilirubin measurements (Yetman et al., 1998). Universally, it is maintained that phototherapy can be stopped once three consecutive measurements have indicated that bilirubin levels are consistently 50 mmol/L below the phototherapy threshold line (Osmond and Northrop, 2022). Further on, it is important to transfer the neonate back into maternal care after phototherapy is stopped and not keep the neonate under supervision inside the neonatal intensive care unit (NICU) to ensure the maternal-neonate bond continues to develop (Queensland Clinical Guidelines, 2022) and prevent the drawbacks as stated above. Lastly, there is a need to maintain vigilance for the re-emergence of jaundice post-discharge. Therefore, keeping these patients under close supervision after stopping interventions is critical (Yetman et al., 1998). This

underscores the pressing need to address potential vulnerabilities in the protocol for managing neonatal jaundice, calling for adherence to guidelines.

2.3 Potential for an AI Coach in Promoting Medical Protocol Adherence

Recent integrations of Artificial Intelligence in healthcare have shown considerable potential to play an important role in promoting adherence to healthcare guidelines by medical professionals. By evaluating real-world data, AI systems can produce insights, improve decision-making and optimise health outcomes and patient safety by providing real-time feedback and decision support to healthcare professionals, ensuring that the care provided aligns with established guidelines and best practices. For example, AI-integrated Clinical Decision Support Systems (CDSS) can recognise real-time actions in the operation room and recommend or alert practitioners when a deviation from guidelines is detected. Such systems don't merely act as passive references, but actively guide and support clinical decisions (Wasylewicz and Scheepers-Hoeks, 2018). Such proactive and dynamic engagements reinforce adherence to guidelines, ultimately driving improved patient outcomes and system efficiency. This is opposed to older systems such as the Modified Early Warning Score (MEWS), a tool used by hospitals, which calculates the mathematical physiological probability of a patient's likelihood of experiencing clinical decline in a short subsequent time interval of an operation, to prevent delay in intervention and promote more vigilance in doctors (AHA, 2023). The implementation of aforementioned AI algorithms-based systems can, however, undoubtedly allow for better support in neonatal medical protocols. The implementation and effect of these AI coaches can be systematically explored by network models, including the simulation of optimal processes, common deviations, and AI interventions which is presented below.

3 Methodology

This section delivers an overview of the network-oriented temporal-causal modelling technique, drawn from the approach outlined by Treur (2016, 2020), which allows the construction of self-models in networks used to run the simulations this article revolves around. Additionally, this section details the motives followed to create the adaptive network model.

3.1 Introduction to the Network-Oriented Self-Modeling Approach

Network-oriented modeling is a technique that can be used for representing, analysing and understanding numerous complex cognitive, biological and social processes that are temporal in nature (Treur, 2016). Through graphically conceptualising a phenomenon, even the space between individual and collective behaviour can be bridged. Adding to this, adaptive reified network models, also known as self-modelling networks, adds in an easy way the concept of adaptation to network models and together create a temporal-causal network which is better suited to model learning and forgetting in a dynamically changing environment.

To elaborate, firstly, a complex real-world process can be represented conceptually as a base-level network which can include connections to different stakeholders in the

process with their own subprocesses as well. Other than graphically representing such networks, they can also be represented in table format through role matrices which is the required format needed as input to simulate the networks in the software environment used. By focusing on self-modelling and creating adaptive networks, which adjust their structure based on internal and external feedback, a powerful environment which allows capturing the dynamic nature of complex scenario settings in the real world such as healthcare and mental models used in protocols to be followed can be created (Treur, 2020).

Mental models operate as essential structures that shape the behavior of individuals. Kim (1997) elucidates these models as encompassing an individual's worldview, inclusive of both implicit and explicit understandings. These models offer a medium for processing and interpreting new information while prescribing the relevance of previously stored knowledge to a given context. These mental models enable individuals to create internal simulations based on specific circumstances. Furthermore, individuals can acquire novel mental models or adapt and improve existing ones, as seen through Piaget's theory, where accommodation (altering one's cognitive concepts based on experiences) and assimilation (integrating experiences into existing cognitive concepts) are keys for learning (Piaget, 1954).

It is possible to add adaptivity to the mental models in the base network by creating a self-model of some of the network characteristics in an added reification (self-modeling) level: the first-order self-modeling or reification level. Reification is defined as “representing something abstract as a material or concrete thing or making something abstract more concrete or real” (Treur, 2020) and is visually placed in a plane above the plane of the base level in the conceptual model. Further on, a second-order self-modeling level can be further added to add adaptivity to the first-order self modeling level as well (metaplasticity) (Treur, 2020). This allows for control of the adaptation in the model (the extent of learning or forgetting) as a whole and creates a three-level architecture to enable control over network characteristics like adaptation speed, the strength of connections, persistence and other fundamental or advanced network attributes. Additionally, in a situation where there exist different pathways in the base network representing a different variation of the processes, context factors exist to exert an external influence to specify which states get activated through having a strong connection while the other context factors have a null weight on their connections (Treur, 2020). In the brain for example, the base-level network could be equated to the neural pathways in the brain which holistically execute a process together by propagation of neuron activations. The first-order self-model could be seen as determining the synaptic transmission between neurons, thereby controlling the strength of the connection. The second-order self-model, would then be seen as modeling the meta-plasticity (plasticity of synaptic plasticity), indicating that the states at this level can modify the extent of (strengthening or weakening) synaptic plasticity (Treur, 2020).

This pattern of self-modelling in a network can be recursively applied to make even higher levels of self-models beyond the second-order level. For example, taking several adaptation levels into account that have been formed during evolution (Treur 2024). And up to five orders of adaptation can be identified for epigenetic effects on mental disorders, e.g., (Kathusing, Samhan, Treur, 2024). For the experimental simulations

conducted during the course of this research, a second-order adaptive self-modeling network is employed for creating simulations of neonatal medical protocols through using internal mental models.

3.2 Design and Specification of Self-Modeling Networks

The fundamental concept in the network-oriented modeling approach is the presence of a labelled graph containing states, which are nodes, where a node X can be connected through a link to another state Y where X has a causal impact on Y at a certain time, and takes into account the strength of connections, the speed of causal effects, and the combination of different incoming connections and the way in which their collective aggregated effect on the state Y . In summary, such characteristics are described as follows (Treur, 2016):

- **Connectivity** characteristics: Connections from a state X to a state Y have a connection weight value, represented by $\omega_{X,Y}$ which represents the strength of the causal impact of state X on state Y through the connection
- **Timing** characteristics: For each state Y , a speed factor represented by η_Y indicates how fast the state is changing upon a causal impact.
- **Aggregation** characteristics: For each state Y , a combination function represented by $c_Y(\cdot)$ determines mathematically how the causal impacts of other states on state Y is combined.

The above concepts are numerically represented and incorporated within related difference equation which represents timing the causal impact (Treur, 2016):

$$Y(t + \Delta t) = Y(t) + \eta_Y [\text{aggimpact}_Y(t) - Y(t)] \Delta t$$

$$\text{where } \text{aggimpact}_Y(t) = c_Y(\omega_{X_1,Y}X_1(t), \dots, \omega_{X_k,Y}X_k(t)) - Y(t)] \Delta t$$

This shows how the causal impact on Y is exerted over time gradually, using speed factor η_Y . This includes single causal impacts represented by $\omega_{X_i,Y}X_i(t)$, where at time t each state X_i with connection to state Y has an impact on Y , using weight $\omega_{X_i,Y}$. Equation (1) collects the aggregated impact of multiple states X_i on Y at time t , which is determined using combination function $c_Y(\cdot)$ (Treur, 2016).

Designing a network model means that at the end the above network characteristics are specified in a systematic manner. A dedicated software environment based on the above network characteristics is utilized to generate the simulation output; see Ch 9 of (Treur, 2020). A visual graph schema of a conceptual self-modeling network model (See Fig. 6) was initially built, and subsequently translated into various role matrices to delineate the network features based on their specific roles. The software environment for the computational work was MATLAB, which took the different role matrices as input to generate the graphical simulation output. These input role matrices include both adaptive and non-adaptive types. Table 1 provides a summary of the descriptions of these matrices, and a comprehensive review of all the role matrices used for each scenario can be found in the Appendix.

Table 1 - Overview of role matrices used in a detailed design and in the software environment.

Role matrix	Description
iv	Initial values: specifies initial values of all state
mb	Base matrix: specifies incoming causal connections for every state
mcwv	Connection weight value matrix: specifies non-adaptive connection weights ω .
mcwa	Connection weight adaptation matrix: specifies adaptive connection weights.
mcfwv	Combination functions weight matrix: specifies function c used for each state.
mcfwa	Combination functions adaptation matrix: specifies adaptative function used for a state.
mcfpv	Combination functions parameter value matrix: specifies function parameter values for non-adaptive parameters.

mcfpa Combination functions parameter adaptation matrix: specifies function parameters for adaptive parameters.

msv Speed value matrix: specifies nonadaptive speed factor η for states.

msa Speed adaptation matrix: specifies adaptive speed values for states

The software environment employs a numerically indexed library of various combination functions to delineate the aggregation traits of the model. The fundamental combination functions are denoted as $bcf_i(p, v)$, utilizing the assigned list of parameters 'p' and values 'v', as detailed in Table 2. In this library, every index number i corresponds to one of over 65 unique mathematical functions that simulate social contagion, learning, synchrony, and homophily, among other concepts (Treur, 2020). Each state Y can be assigned only one combination function; however, the complete model may incorporate multiple different functions for different states. For every state Y , the weighted mean is computed using the combination function weights that are housed in the **mcfwv** and **mcfwa** matrices with the following calculation:

$$\mathbf{c}_Y(\pi_{1,1,Y}, \pi_{2,1,Y}, \dots, \pi_{1,m,Y}, \pi_{2,m,Y}, \dots, V_1, \dots, V_k) = \frac{\gamma_{1,Y} bcf_1(\pi_{1,1,Y}, \pi_{2,1,Y}, V_1, \dots, V_k) + \dots + \gamma_{m,Y} bcf_m(\pi_{1,m,Y}, \pi_{2,m,Y}, V_1, \dots, V_k)}{\gamma_{1,Y} + \dots + \gamma_{m,Y}}$$

Table 2 - Overview of combination functions and their parameters selected to model the simulations in this article.

Name	Notation	Formula	Parameters
Identity	$\text{id}(V_1, \dots, V_k)$	V	No Parameter
Advanced Logistic Sum	$\text{alogistic}\sigma, \tau(V_1, \dots, V_k)$	$\left[\frac{1}{1+e^{-\sigma(V_1+\dots+V_k-\tau)}} - \frac{1}{1+e^{\sigma\tau}} \right] (1 + e^{-\sigma\tau})$	Steepness σ Excitability threshold τ
Max Hebbian Learning Composition	$\text{maxhebb}\mu(V_1, \dots, V_k)$	$\max(V_1 V_2 (1 - V_3) + \mu V_3, V_4, \dots, V_k)$	Persistence factor μ

In (Treur, 2021a; Hendrikse, Treur, Koole, 2023), it is shown that any smooth dynamical system has a canonical representation as a temporal-causal network and any smooth (multi-order) adaptive dynamical system has a canonical representation as a (multi-order) self-modeling temporal-causal network. Therefore, compared to adaptive dynamical systems in general, the network-oriented modeling approach used here does not introduce any fundamental limitations concerning what it can model. This has also been confirmed by applications to case studies in practice, e.g., (Treur and Van Ments, 2022; Canbaloglu, Treur, Wiewiora, 2023).

To assimilate the above and implement the self-modelling network in the context of this article, a model of the healthcare guidelines has to be made for the real-world process. In addition to the process of the guidelines in the real world, the inclusion of a health care practitioner (doctor) to execute the protocol and showcase the interactions between the process being followed is needed. This inclusion will be done through the creation of an internal mental model of the health care practitioner at the base level, to show their thinking process which acts as the preparation before translating into their actions towards the extent they follow the medical protocol in the real world. The healthcare practitioner will be abbreviated to “HCP” in further explanations and in representing the mental states and the mental model of the healthcare practitioner in the conceptual model. Moreover, this paper aims to model and determine the effectiveness of a virtual Artificial Intelligence Coach in fostering adherence of the HCP in following the medical protocol. To observe this, a separate mental model of the AI coach is included alongside that of the HCP at the base level of the conceptual model. The AI coach will be abbreviated to AIC in further explanations and in representing the mental states and the mental model of the AI coach in the conceptual model. The AIC should become aware of any deviations in the execution of the medical protocol and may provide support to the HCP by communicating to alert the HCP of the deviation so as to correct them. Both the HCP and AIC mental models contain a fully identical rendition of the medical protocol to signify that both entities in principle possess the complete knowledge representation of the medical protocol in order to be able to carry out their tasks.

3.3 Modelling Decisions and Methods

In the context of this article, three scenarios were modelled using the self-modelling approach to show how an artificial intelligence coach (AIC) can intervene to encourage adherence to guidelines. This will be addressed for both the neonatal medical protocols for the treatment of Neonatal Hypoglycemia and Neonatal Hyperbilirubemia. The first scenario simulates the medical protocol being followed by the health care practitioner HCP with a perfect mental model. This means that the HCP does not commit any mistakes while following the protocol and receives input from the real world, comprehends, stores it perfectly, and effectuates the corresponding actions needed to follow the protocol without any deviations, delays or mistakes. It is important to note that due to prior experience and education the doctor knows what steps to take when exposed to contextual stimuli from the real-world states such as specific measurement readings. The second scenario deals with simulating the HCP following the medical protocols with an imperfect mental model. This means that the HCP makes mistakes while following the protocol and does not effectuate the actions in an ideal manner in the real world. The third scenario will in two parts simulate the HCP following the medical protocol with the same perfect and imperfect mental model as Scenario 1 and 2, however an artificial intelligence coach (AIC) with perfect mental model is introduced to intervene and prevent any deviations when required. It is important to note that unlike the doctor the AIC does not have prior experience and education and does not know the steps that will be taken or followed by the doctor when exposed to contextual stimuli from real-world states such as specific measurement readings and thus needs constant inputs from all world-states to become aware of which steps in the protocol, thus which world-states in the model, are being executed and to what extent have they been carried out.

Based on the background literature described in Section 2, a conceptual model (see Figs. 1 and 2 below) for each medical protocol was built to establish the framework upon which the final self-modeling network would be simulated. The protocol for the prevention of neonatal hypoglycemia was modelled conceptually following the medical protocol flowchart provided by the Dutch National Union of Child Medicine (NVK, 2020). The protocol for prevention of neonatal hyperbilirubinemia was modelled conceptually following the medical protocol flow chart provided by University Hospital Bristol and Weston (Osmond and Northrop, 2022). The concepts discussed in the background literature and in these medical protocols were used in the designed computational network model that was simulated using the software described in Section 3.2.

The conceptualised model envisages a procedure in line with the established protocol which models different paths to be followed depending on different situations in the real world concerning the physical conditions of the neonate. Sections 4.2 and 4.5 describe the specific pathway that was followed in the medical protocols during the simulation. There are crucial measuring states incorporated within the protocols which branch into various measurement readings and give rise to following a different pathway in the protocol, tailored to treat the specific circumstances corresponding to the measurements. To capture a specific circumstantial situation related to the unique conditions of the neonate, context factors are used to allow for the activation of certain

measurements and thus their consequent pathways and make sure that the states exclusively included in the pathways which arise from other measurements are not activated as that would be illogical and contradictory. Setting the context factors' initial value and the incoming connections to the contextual nodes at '1', activates the measurements and their subsequent pathway while setting the value and weights to '0' ensures that those states are not activated. For instance, if a baby does not have small gestational age as a risk factor but has large gestational age (can never have both), then the context factors for small gestational age will be set at '0' and '1' will be set for the context factor of the latter. In the conceptualised model of this article, the context factors and thus the pathway taken to treat the most severe conditions and symptoms of the Hypoglycemia and Hyperbilirubinemia has been chosen (nhs and sJ respectively).

As highlighted above, the combination functions used for aggregations in the states of the conceptual model have been chosen to be **id**, **alogistic** and **maxhebb**. The **alogistic** function is used for states on the base level which have incoming connections to them. For states with no incoming connections such as context factors and the initial state of neonate being born all have combination function set to **id** meaning they need no aggregations done to them. For the first-order adaptation states, the max Hebbian learning composition function was assigned. This combines the max function with an adaptation principle (Hebb, 1943) which in a simplified form follows the motto "neurons that fire together, wire together". Specifically, the function **maxhebb** allows for the adaption of the connection weights (strength) between the states in the base network. This is critical to the framework of the model as it plays a pivotal role in the modelling of deviations and learning in the mental state of the HCP as described below.

Hebbian learning is applied to the (middle level) first-order self-model states $\mathbf{W}_{X,Y}$ that represent the adaptation of the agents' internal mental models. These self-model states and their dynamics highlight a mental model's learning process. Hebbian learning has one parameter μ called persistence factor. A persistence factor represents the fraction that persists each time unit. For example, $\mu = 0.95$ means that for each time unit 5 percent of a learnt value of a first-order self-model $\mathbf{W}$ -state can be lost. It is also possible that the persistence factor of $\mathbf{W}_{X,Y}$ is adaptive too, represented by a second-order self-model $\mathbf{M}_{\mathbf{W}}$ -state $\mathbf{M}_{\mathbf{W}_{X,Y}}$. In the context of this article, the second-order $\mathbf{M}_{\mathbf{W}}$ -states are influenced by the context state in the base level. They control the adaptation of mental models (metaplasticity), for example stressful circumstances may make the level of $\mathbf{M}_{\mathbf{W}_{X,Y}}$ lower which means more forgetting.

Note that **all** the actions of the HCP (and thus the prerequisite preparation states in the mental model of the HCP due to to which those actions resulted) will be monitored by the AI Coach to ensure that not a single deviation takes place. In the context of this paper to show that the AIC can display behaviour which better induces adherence to guidelines in different medical protocols, specific vulnerabilities for each protocol which were described in the background literature (Section 2) have been chosen to be the parts of the protocol where the HCP makes a mistake when it has an imperfect mental model so that AIC has to notice that and intervene.

The methodology used to model deviations for simulating Scenario 2 in both medical protocols is as follows. To show that the HCP has not followed a certain step in the

medical protocol correctly, there has to be a flaw in their thinking within their mental model which has caused this flaw to happen in the real world. In a perfect mental model, to show that all the connections in the mind are perfect, the weight of the connections is set to 1. Note that any connection in any of the mental models ranges from connection weights 0 (no connection) to 1 (perfect connection). However, in an imperfect mental model the connection weight is set to a very low level which signifies a flaw or omission in the mental model. In the instance of the models created in this paper, the mental states with flawed connections have been modelled with adaptations to show learning and awareness by the HCP that the mistake has been committed and further adaptation of the persistence (second-order) to show the level of forgetfulness of this learning (from the first order). The connections of the base-level mental model states represent the strengths and vulnerabilities in the protocol. These are represented by the first-order **W**-states representing the weights of connections between the base-level states. For vulnerable cases, they have been set to have a low connection weight to signify having a flaw in the thinking process of the person. Since these **W**-states determine the actual connection weight strength between the states, the situation can change dynamically by learning or forgetting.

The AIC constantly monitors the changes in the actions taken in the real world and recognises whether there has been a shortcoming in the presence and execution of a certain step in a medical protocol corresponding to a state in the mental model. It is assumed here that the monitoring of the real world done by the AIC is carried out perfectly, with the weights of the monitoring connection from the world-states to the AIC mental model always having a value of 1. If there is a state in the world which has to be executed by the HCP and has not been executed ideally, the corresponding state in the AIC mental model signifies to the HCP (and thereby specifically to the HCP mental model) that a mistake has been committed and the specific weak connection should be strengthened to attain a more ideal thought process which translates into performing the protocol with more accuracy and adherence. This communication from AIC to HCP has been modelled by linking an arrow from the vulnerable state in the AIC mental model to the 1st-order self-model **W**-state representing the weak connection for the vulnerable state in the HCP mental model. This arrow has a strong positive weight connection of 1 and increases the strength of the 1st-order **W**-state, translating into a positive adaptation and learning and thus a better functioning of the HCP mental model. Note that the AIC only provides such support when the HCP mental model is flawed, and the AIC detects those flaws: these actions are suppressed when no mistake is committed.

Table 3 - Legend for Schematic Elements in the Adaptive Network Model

Component	Description
Purple Node	Second Order Self-Modelling state, M , representing persistence μ
Purple Level	Second Order reification level
Blue Node	First Order Self-Modelling state, W , representing connection weight ω
Blue Level	First Order reification level
Pink Node	World States in real world
Pink Level	Base Level
Yellow Node	HCP mental model state
Beige Level	HCP Mental Model
Beige Node	AIC mental model state
Yellow Level	AIC Mental Model
Red Arrow	Incoming connections from real world states to mental model states
Green Arrow	Outgoing connection from mental model states
Blue Arrow	Upward connections from base level states to first order self-model state

Pink Arrow	Downward connections between reification levels
Black Arrow	Connections between real world states.

The granularity of the states listed in Table 3 make them ideal for network modelling. Through the application of the various previously tabled combination functions for different linear and non-linear connections a complex simulation can be created for various scenarios that reproduce real-life conditions.

It is also important to detail the starting points of various states. This is modelled by initial values which in simulation graphs signify a value on the vertical axis corresponding to that state's activation level in the simulation. Specifically, while most states start a 0 level, depending on the scenario of the simulation the **W**-states and the **M_W**-persistence states can start at an initial higher level corresponding to the mental model the HCP has. In an ideal scenario for example, the doctor has an ideal mental model and start with a high level of knowledge which was previously learnt by him and which he remembers and also a strong level of persistence as to not forget that knowledge. However, in a scenario with a weak mental model, the initial values of the adaptation **W**-states are the opposite. In the models, while the **W**-states of a weak mental model have 0.1 as an initial value, the **M_W**-states have a null initial value while in a strong mental model scenario the **W**-states have an initial value of 0.9 while the **M_W**-states have 0.8 as their initial value, respectively. Additionally, to start the simulation the first node where the neonate is born in real life has to have an initial value of 1, to show the occurrence of the event and lastly context factors also have an initial value of 1 or 0 depending on their presence in the real world. Further on, while due to convention this is not conveyed in the graphical picture of the adaptive networks, all **W**-state have a self-connection to themselves to reinforce persistence and be able to improve due to their learning.

Lastly, it is important to touch on the topic of the specific values of parameters which have been chosen throughout all scenarios of the conceptual models. The speed of all states, signifying their rate of change, has been kept at 0.5 to ensure control of a state not changing its value unrealistically relative to other states, while context factor states have a speed of 0, only an initial value of 1 to indicate their presence as it is not a dynamic factor which can change its level of existence in a real-life scenario. Monitoring states have a speed of 0.05 to ensure there is a time delay between a fault taking place in the protocol, and then being detected by the monitoring states after which their signals allow the protocol to be corrected gradually. **W**-states, however, have a speed value of 0.2 to ensure changes to the level of learning happen gradually, reflecting how evolution of thinking in mental models happens in real life and not drastically. Further on steepness and threshold of any present state in the protocol using the **alogistic** function in the model is kept at 5 and 0.5 respectively. Monitoring states

are allocated steepness 30 and threshold 0.4 respectively to display after being dormant, a sudden and steep rise in presence the moment they find a fault arising in the protocol. The same reasoning is given to the vulnerable mental model states which have downward effectuating connections incoming from **W**-states and represent where the flaw in the mental model and the mental process of the HCP occurs, further affecting the that corresponding part of the protocol in the real-world states. These mental model states thus need to reflect a similar pattern of sudden change as the monitoring states, which is why they also have a higher steepness and lower threshold value of 10 and 0.3 respectively.

4 Design of the Adaptive Network Models

This section depicts the conceptual models of the self-modelling adaptive networks of the Neonatal Hypoglycemia and neonatal hyperbilirubinemia medical treatment protocols and their corresponding legends for giving an overview of the states shown in the models.

4.1 Neonatal Hypoglycemia Model

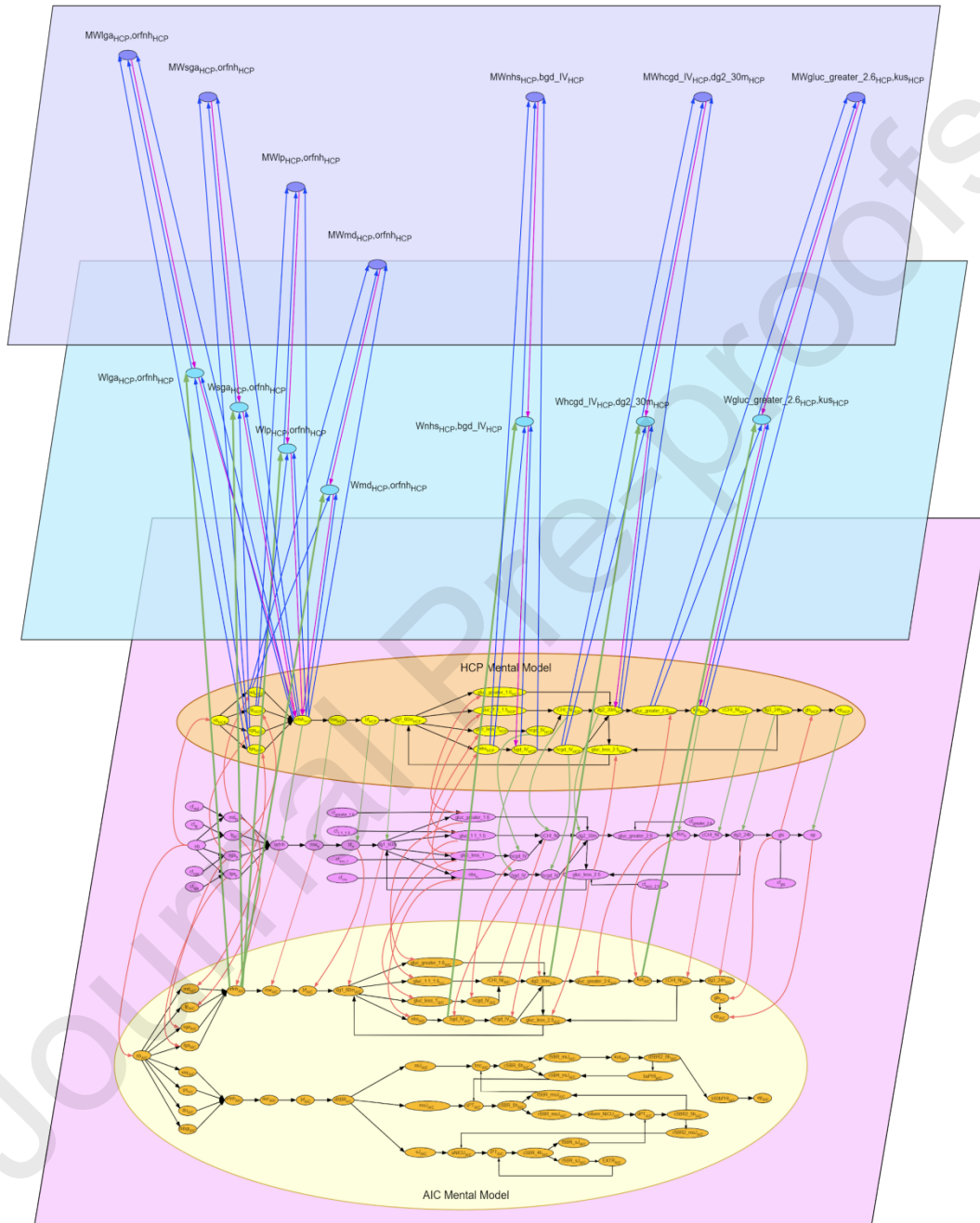
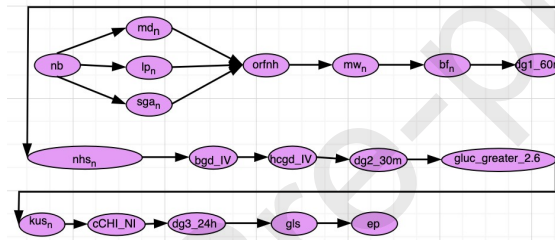


Fig. 1 - Final Conceptual model depicting medical protocol to treat

Neonatal Hypoglycemia including the presence of HCP and AIC mental models.

To show the base-level network more clearly and without any interference by connections to the second order states from the base-level mental model, the following figure presents the base level mental model independently. Note that zooming into the following figure allows more clarity in observing what the names of each of the states are.

To model the treatment as a real-life situation, a certain path trajectory of the conceptual model of the treatment has to be followed. As stated earlier in, the worst scenario is followed and in the context of the above treatment, this corresponds to the following schema:



When the neonate is born, it is small for its gestational age, it is born to a mother who has diabetes, and it is born with late prematurity. These are major risk factors for neonatal hypoglycemia and are observed by the doctor. To treat this, the doctor measures weight and gives the baby for breastfeeding to ensure glucose levels stay normal. The glucose level is then determined after 60 minutes and shown to be extremely low showing symptoms of neonatal hypoglycemia. An intravenous bolus glucose dose is administered as a drastic measure, immediately followed by a high continuous glucose dose. The glucose level is determined the second time and shown to have improved to above 2.6 mmol/L. The neonate is kept under supervision and given carbohydrate food, until the third glucose measurement after 24 hours shows glucose levels have stabilised and thus the protocol is ended.

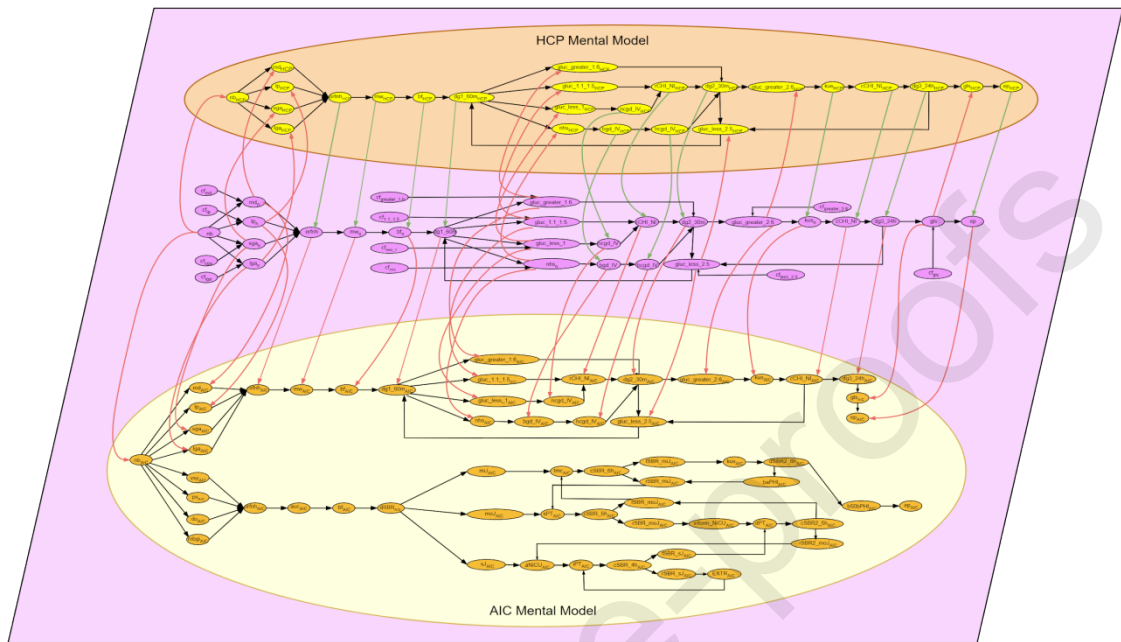


Fig. 2. - Base network level for the conceptual model of the medical protocol for the treatment of Neonatal Hypoglycemia

4.2. Legend For Neonatal Hypoglycemia Model

Table 4 - Overview of states used in Neonatal Hypoglycemia conceptual model alongside their corresponding description

State nr	State name	Explanation	Level
X1	nb	Neonate Born: Otherwise healthy neonates (≤ 24 hours old) with a gestational age > 35 weeks and birth weight > 2000 grams	

X2	md _n	Maternal Diabetes concerning mother of neonate
X3	lp _n	Late Prematurity of Neonate
X4	sga _n	Neonate has weight Small for Gestational Age
X5	lga _n	Neonate has weight Large for Gestational Age
X6	orfnh	Observe risk factors in neonate for neonatal hypoglycemia
X7	mw _n	Measure weight of the neonate for future reference needed for administering certain amount of glucose to the neonate dependant on the body weight
X8	bf _n	Begin breastfeeding neonate within 1 hour of birth
X9	dg1_60m	Determine glucose concentration 60 minutes after birth; after the first feeding. Target value $\geq$ 2.0 mmol/L

X10	gluc_greater_1.6	Glucose concentration measured is greater than or equal to 1.6 mmol/L	
X11	gluc_1.1_1.5	Glucose concentration measured is between 1.1 mmol/L and 1.5 mmol/L	
X12	gluc_less_1	Glucose concentration measured is less than or equal to 1 mmol/L	
X13	nhs _n	Neonatal hypoglycemia symptoms shown due to extremely low glucose concentration	
X14	ncgd_IV	Normal continuous glucose dose administered intravenously.	
X15	bgd_IV	bolus (push) glucose dose administered intravenously, in case of severe hypoglycemia	Base Level - World States
X16	rCHI_NI	Raise Carbohydrate Intake through a Nutritional Intervention	
X17	hcgd_IV	High continuous glucose dose administered intravenously, right after	

		the bolus glucose dose in case of severe hypoglycemia.
X18	dg2_30m	Determine glucose concentration 30 minutes after initial interventions. Target value ≥ 2.6 mmol/L
X19	gluc_less_2.5	Glucose concentration measured is less than or equal to 2.5 mmol/L
X20	gluc_greater_2.6	Glucose concentration measured is greater than or equal to 2.6 mmol/L
X21	kus _n	Keep neonate Under Supervision for 24 hours to further see whether glucose level remains stable. Remove all other interventions
X22	cCHI_NI	Continue Carbohydrate Intake through a Nutritional Intervention.
X23	dg3_24h	Determine glucose concentration 24 hours after supervision. Target value ≥ 2.6 mmol/L
X24	gl _s	Glucose Level is Stable and above 2.6 mmol/L

X25	ep	End Protocol	
X26	cf_{md}	Context factor state md_n	
X27	cf_{lp}	Context factor state lp_n	
X28	cf_{sga}	Context factor state sga_n	
X29	cf_{lga}	Context factor state lga_n	Base Level
			-
X30	$cf_{greater_1.6}$	Context factor state $gluc_greater_1.6$	Context Factor
			States
X31	$cf_{1.1_1.5}$	Context factor state $gluc_1.1_1.5$	
X32	cf_{less_1}	Context factor state $gluc_less_1$	
X33	cf_{nhs}	Context factor state nhs_n	
X34	$cf_{less_2.5}$	Context factor state $gluc_less_2.5$	
X35	$cf_{greater_2.6}$	Context factor state $gluc_greater_2.6$	

X36	cf_{gls}	Context factor state gls_n
X37	nb_{HCP}	HCP mental model state for registering that Neonate is Born
X38	md_{HCP}	HCP mental model state for registering that neonates mother has Diabetes
X39	lp_{HCP}	HCP mental model state for registering that Neonate has had Late Prematurity in pregnancy
X40	sga_{HCP}	HCP mental model state for registering that Neonate has weight Small for Gestational Age
X41	lga_{HCP}	HCP mental model state for registering that Neonate has weight Large for Gestational Age
X42	$orfnh_{HCP}$	HCP mental model state for realising and observing the risk factors for neonatal hypoglycemia in the neonate

X43	mw_{HCP}	HCP mental model state for Measuring the weight of the neonate
X44	bf_{HCP}	HCP mental model state to initiate breastfeeding procedure neonate within 1 hour of birth
X45	dgl_60m_{HCP}	HCP mental model state to Determine glucose concentration 60 minutes after birth; after the first feeding. Target value ≥ 2.0 mmol/L
X46	$gluc_greater_1.6_{HCP}$	HCP mental model state to register that Glucose concentration measured is greater than or equal to 1.6 mmol/L
X47	$gluc_1.1_1.5_{HCP}$	HCP mental model state to register that Glucose concentration measured is between 1.1 mmol/L and 1.5 mmol/L
X48	$gluc_less_1_{HCP}$	HCP mental model state to register that Glucose concentration measured is less than or equal to 1 mmol/L
X49	nhs_{HCP}	HCP mental model state to register that Neonatal hypoglycemia symptoms

		shown (due to extremely low glucose concentration)	
X50	ncgd_IV _{HCP}	HCP mental model state to administer Normal continuous glucose dose intravenously.	
X51	bgd_IV _{HCP}	HCP mental model state to administer bolus (push) glucose dose intravenously, in case of severe hypoglycemia	
X52	rCHI_NI _{HCP}	HCP mental model state to effectuate Raising Carbohydrate Intake through a Nutritional Intervention	
			Base Level
			-
X53	hcgd_IV _{HCP}	HCP mental model state to administer a High continuous glucose dose intravenously, in case of severe hypoglycemia.	HCP Mental Model
X54	dg2_30m _{HCP}	HCP mental model state to Determine glucose concentration 30 minutes after initial interventions.	

		Target value ≥ 2.6 mmol/L
X55	gluc_less_2.5 _{HCP}	HCP mental model state to register that Glucose concentration was measured less than or equal to 2.5 mmol/L
X56	gluc_greater_2.6 _{HCP}	HCP mental model state to register that Glucose concentration was measured greater than or equal to 2.6 mmol/L
X57	kus _{HCP}	HCP mental model state to effectuate Keep neonate Under Supervision for 24 hours to further see whether glucose level remains stable. All other interventions removed
X58	cCHI_NI _{HCP}	HCP mental model state to effectuate Continuing Carbohydrate Intake through a Nutritional Intervention.
X59	dg3_24h _{HCP}	HCP mental model state to Determine glucose concentration 24 hours after supervision. Target value ≥ 2.6 mmol/L
X60	gl _{sHCP}	HCP mental model state to register that Glucose

Level is Stable and above 2.6 mmol/L		
X61	ep_{HCP}	HCP mental model state to affectuate Ending the Protocol
X62	$Wmd_{HCP, orfnh_{HCP}}$	Reified representation state for connection weight $\omega md_{HCP, orfnh_{HCP}}$
X63	$Wlp_{HCP, orfnh_{HCP}}$	Reified representation state for connection weight $\omega lp_{HCP, orfnh_{HCP}}$
X64	$Wsga_{HCP, orfnh_{HCP}}$	Reified representation state for connection weight $\omega sga_{HCP, orfnh_{HCP}}$
X65	$Wlga_{HCP, orfnh_{HCP}}$	Reified representation state for connection weight $\omega lga_{HCP, orfnh_{HCP}}$
X66	$Wnhs_{HCP, bgd_IV_{HCP}}$	Reified representation state for connection

First

Reification

Level

		$\text{weight}_{\omega\text{h}_{\text{HCP}},\text{bgd_IV}_{\text{HCP}}}$	
X67	$\text{W}_{\text{hcgd_IV}_{\text{HCP}},\text{dg2_30m}_{\text{HCP}}}$	Reified representation state for connection $\text{weight}_{\omega\text{hcgd_IV}_{\text{HCP}},\text{dg2_30m}_{\text{HCP}}}$	
X68	$\text{W}_{\text{gluc_greater_2.6}_{\text{HCP}},\text{kus}_{\text{HC}}\text{p}}$	Reified representation state for connection $\text{weight}_{\omega\text{gluc_greater_2.6}_{\text{HCP}},\text{kus}_{\text{HC}}\text{p}}$	
X69	$\text{MW}_{\text{md}_{\text{HCP}},\text{orfnh}_{\text{HCP}}}$	Reified representation state for persistence factor $\mu\text{W}_{\text{md}_{\text{HCP}},\text{orfnh}_{\text{HCP}}}$ for reified representation state $\text{W}_{\text{md}_{\text{HCP}},\text{orfnh}_{\text{HCP}}}$	
X70	$\text{MW}_{\text{lp}_{\text{HCP}},\text{orfnh}_{\text{HCP}}}$	Reified representation state for persistence factor	Second Reification Level

		$\mu W_{lp_{HCP}, orf_{nh_{HCP}}}$ for reified representation state $W_{lp_{HCP}, orf_{nh_{HCP}}}$
X71	$MW_{sga_{HCP}, orf_{nh_{HCP}}}$	Reified representation state for persistence factor $\mu W_{sga_{HCP}, orf_{nh_{HCP}}}$ for reified representation state $W_{sga_{HCP}, orf_{nh_{HCP}}}$
X72	$MW_{lga_{HCP}, orf_{nh_{HCP}}}$	Reified representation state for persistence factor $\mu W_{lga_{HCP}, orf_{nh_{HCP}}}$ for reified representation state $W_{lga_{HCP}, orf_{nh_{HCP}}}$
X73	$MW_{nhs_{HCP}, bgd_IV_{HCP}}$	Reified representation state for persistence factor $\mu W_{nhs_{HCP}, bgd_IV_{HCP}}$ for reified representation state $W_{nhs_{HCP}, bgd_IV_{HCP}}$
X74		Reified representation state for persistence factor

	$MW_{hcgd_IV_{HCP},dg2_30m_H}$ CP	$\mu W_{hcgd_IV_{HCP},dg2_30m_H}$ CP for reified representation state $W_{hcgd_IV_{HCP},dg2_30m_H}$ P
X75	$MW_{gluc_greater_2.6_{HCP},kus}$ HCP	Reified representation state for persistence factor $\mu W_{gluc_greater_2.6_{HCP},kus}$ HCP for reified representation state $W_{gluc_greater_2.6_{HCP},kus}$ CP
X76	nb_{AIC}	AIC mental model state for registering that Neonate is Born
X77	md_{AIC}	AIC mental model state for registering that neonates mother has Diabetes
X78	lp_{AIC}	AIC mental model state for registering that Neonate has had Late Prematurity in pregnancy
X79	sga_{AIC}	AIC mental model state for registering that Neonate has weight Small for Gestational Age

X80	lga_{AIC}	AIC mental model state for registering that Neonate has weight Large for Gestational Age	
X81	$orfnh_{AIC}$	AIC mental model state for realising and observing the risk factors for neonatal hypoglycemia in the neonate	
X82	mw_{AIC}	AIC mental model state for monitoring that the weight of the neonate has been measured	
X83	bf_{AIC}	AIC mental model state for monitoring that the breastfeeding procedure neonate happens within 1 hour of birth	
X84	dgl_60m_{AIC}	AIC mental model state to monitor that the glucose concentration has been determined 60 minutes after birth; after the first feeding.	Base Level - AIC Mental Model
X85	$gluc_greater_1.6_{AIC}$	AIC mental model state to register that Glucose concentration measured is	

		greater than or equal to 1.6 mmol/L
X86	gluc_1.1_1.5 _{AIC}	AIC mental model state to register that Glucose concentration measured is between 1.1 mmol/L and 1.5 mmol/L
X87	gluc_less_1 _{AIC}	AIC mental model state to register that Glucose concentration measured is less than or equal to 1 mmol/L
X88	nhs _{AIC}	AIC mental model state to register that Neonatal hypoglycemia symptoms shown (due to extremely low glucose concentration)
X89	ncgd_IV _{AIC}	AIC mental model state for monitoring that the Normal continuous glucose dose intravenously has been administered.
X90	bgd_IV _{AIC}	AIC mental model state for monitoring that the bolus (push) glucose dose has been administered intravenously, in case of severe hypoglycemia

X91	$rCHI_NI_{AIC}$	AIC mental model state for monitoring that the neonate's Carbohydrate Intake has been raised through a Nutritional Intervention
X92	$hcgd_IV_{AIC}$	AIC mental model state for monitoring that a High continuous glucose dose has been administered intravenously, in case of severe hypoglycemia.
X93	$dg2_30m_{AIC}$	AIC mental model state for monitoring that glucose concentration 30 minutes after initial interventions has been determined. Target value ≥ 2.6 mmol/L
X94	$gluc_less_2.5_{AIC}$	AIC mental model state to register that Glucose concentration was measured less than or equal to 2.5 mmol/L
X95	$gluc_greater_2.6_{AIC}$	AIC mental model state to register that Glucose concentration was measured greater than or equal to 2.6 mmol/L

X96	ku_{AIC}	AIC mental model state for monitoring that neonate Kept Under Supervision for 24 hours to further see whether glucose level remains stable. All other interventions removed
X97	$cCHI_NI_{AIC}$	AIC mental model state for monitoring Continuing Carbohydrate Intake through a Nutritional Intervention for the neonate.
X98	$dg3_24h_{AIC}$	AIC mental model state for monitoring glucose concentration 24 hours after supervision. Target value ≥ 2.6 mmol/L
X99	gls_{AIC}	AIC mental model state to register that Glucose Level is Stable and above 2.6 mmol/L
X100	ep_{AIC}	AIC mental model state for monitoring that the Protocol is ended
X101		

	monitor_md _{orfnh}	Monitor state for AIC mental model for monitoring world state orfnh with connection from state md	
X102	monitor_lp _{orfnh}	Monitor state for AIC mental model for monitoring world state orfnh with connection from state lp	
X103	monitor_sga _{orfnh}	Monitor state for AIC mental model for monitoring world state orfnh with connection from state sga	First Reification Level
X104	monitor_lga _{orfnh}	Monitor state for AIC mental model for monitoring world state orfnh with connection from state lga	
X105	monitor _{bgd_IV}	Monitor state for AIC mental model for monitoring world state bgd_IV	

X106	$\text{monitor}_{\text{dg2_30m}}$	Monitor state for AIC mental model for monitoring world state dg2_30m
X107	$\text{monitor}_{\text{kus}}$	Monitor state for AIC mental model for monitoring world state kus

4.3 Neonatal Hyperbilirubinemia Model

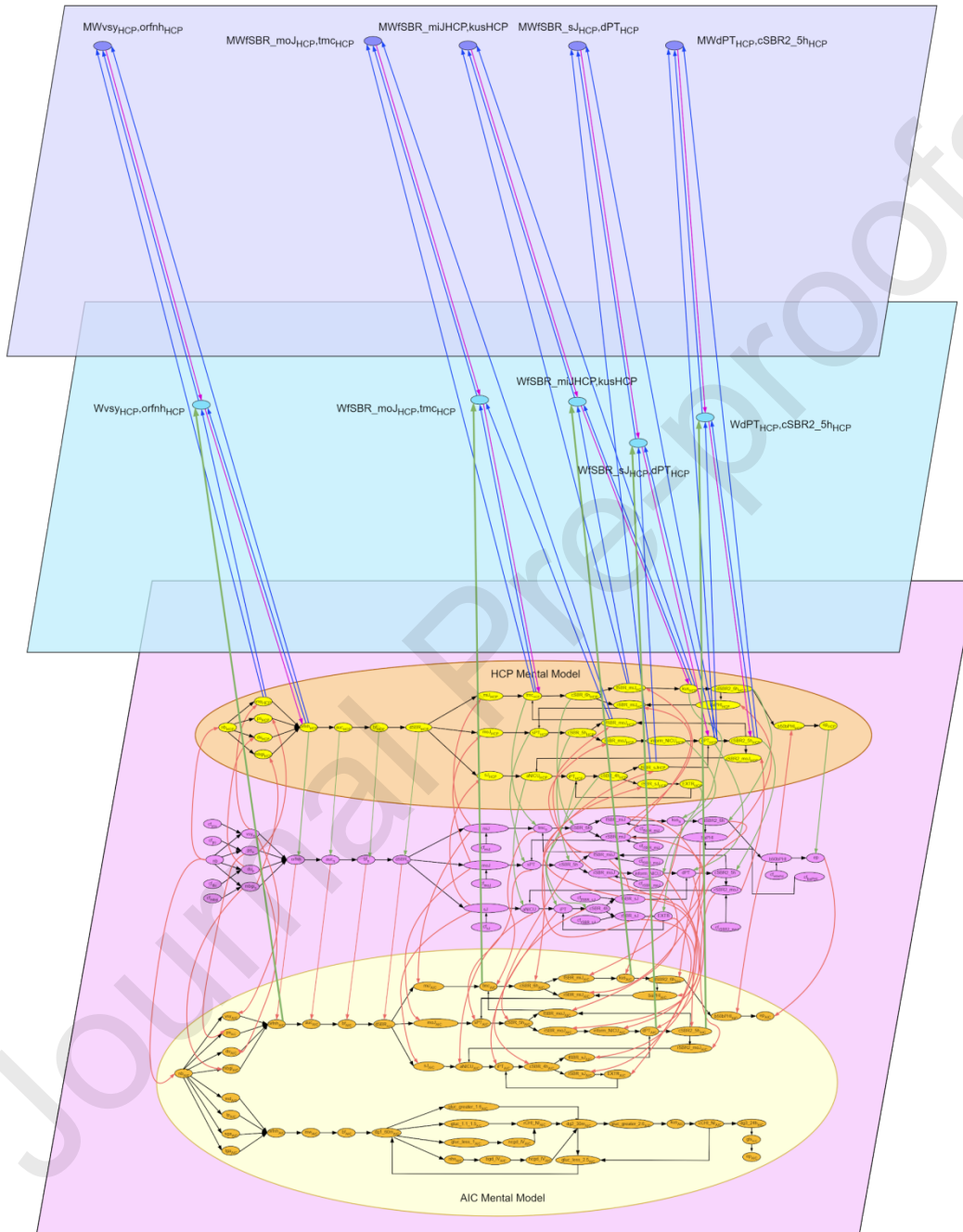


Fig. 3 - Final Conceptual model depicting medical protocol for treatment of Neonatal Hyperbilirubinemia including the presence of HCP and AIC mental models.

To show the base-level network more clearly and without any interference by connections to the second order states from the base-level mental model, the following figure presents the base level mental model independently.

Note that zooming into the following figure allows more clarity in observing what the name of each of the states is.

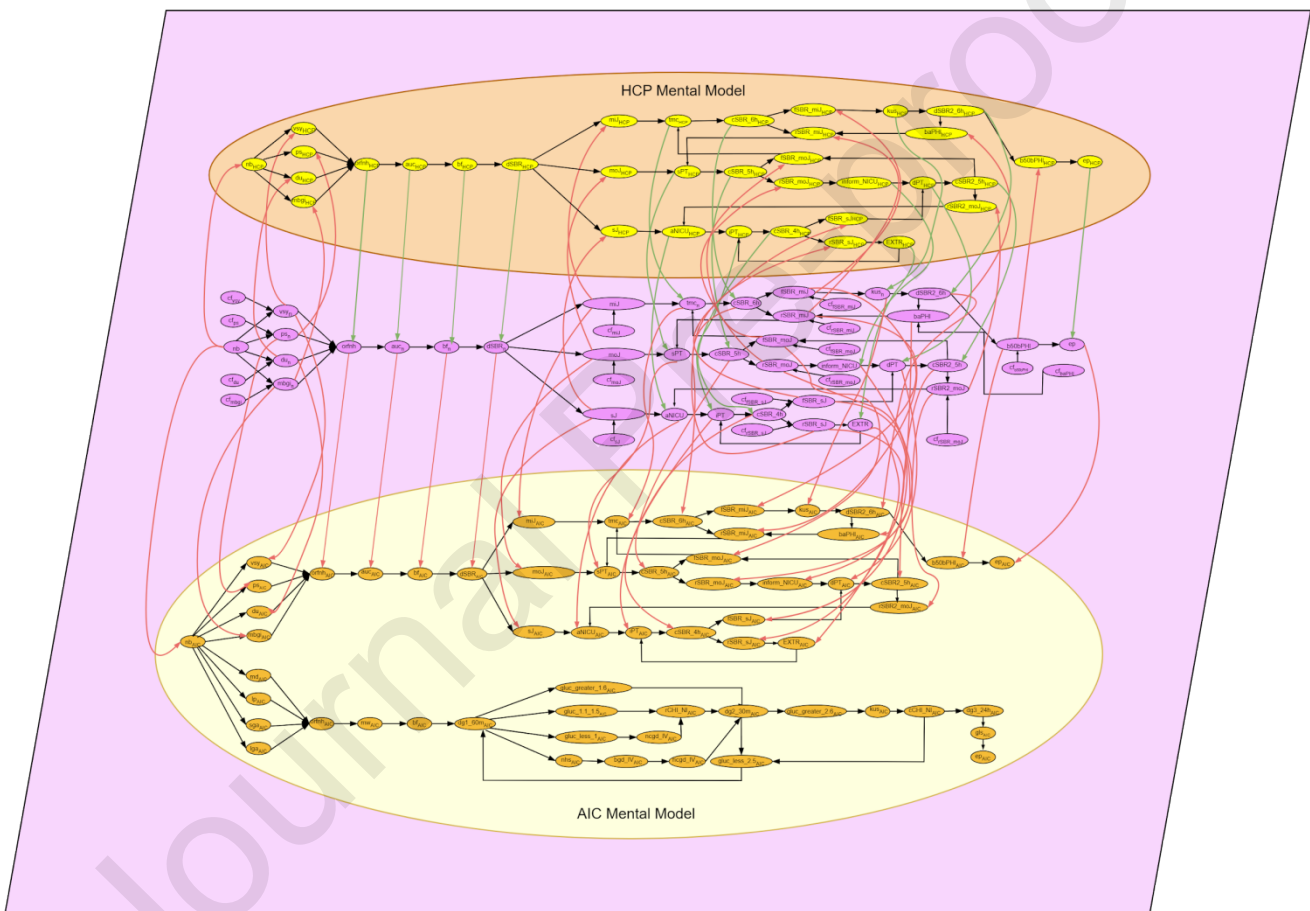
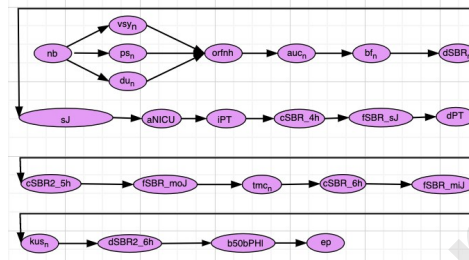


Fig. 4 - Base network level for the conceptual model of the medical protocol for the treatment of Neonatal Hyperbilirubinemia.

In order to model the treatment as a real-world situation in the software environment, a certain path trajectory of the conceptual model of the treatment has to be followed. As stated earlier in the report, the worst scenario is followed and in the context of the above treatment, this corresponds to the following schema:



When the neonate is born, it has yellow skin, and discharges pale stool and dark urine. These are major risk factors for neonatal hyperbilirubinemia and are observed by the doctor. To treat this, the doctor assesses the underlying cause for jaundice and gives the baby for breastfeeding. The serum bilirubin concentration is then determined to be extremely high representing severe jaundice. It is then admitted to the neonatal intensive care unit and undergoes intensive phototherapy as a drastic measure. The SBR level is checked after 4 hours and is shown to be falling. Thereafter, the neonate undergoes double phototherapy. The SBR level is checked after 5 hours and shown to have fallen. The neonate is therefore transferred back to the mother, where SBR level check after 6 hours and still shown to be falling. With neonate now below mild jaundice, it is kept under supervision and SBR level checked once again after 6 hours. Since it has fallen further to below 50 mg/l of the phototherapy line, the protocol is ended.

4.4 Legend for the Neonatal Hyperbilirubinemia Model

Table 5 - Overview of states used in Neonatal Hyperbilirubinemia conceptual model alongside their corresponding description.

State nr	State name	Explanation	Level
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X1	nb	Neonate Born
X2	vsy _n	Neonate Skin is Visually Yellow
X3	ps _n	Neonate has pale stools
X4	du _n	Neonate has dark urine
X5	mbgi _n	Neonate has incompatibility with maternal blood type
X6	orfnh	Observe risk factors in neonate for neonatal hyperbilirubinemia
X7	auc _n	Assess underlying causes for neonatal hyperbilirubinemia, needed for future purposes if initial treatments do not work.
X8	bf _n	Begin breastfeeding neonate within 1 hour of birth
X9	dSBR _n	Determine serum bilirubin concentration of neonate in the blood by taking a blood sample

X10	miJ	Neonate has Mild Jaundice	
X11	moJ	Neonate has Moderate Jaundice	
X12	sJ	Neonate has Severe Jaundice	
X13	tmc _n	Transfer neonate to mother	
X14	sPT	Single Phototherapy done on neonates	
X15	aNICU	Neonate admitted to Neonatal Intensive care unit	Base Level -
X16	cSBR_6h	Check Serum Bilirubin levels in Neonate after 6 hours	World States
X17	cSBR_5h	Check Serum Bilirubin levels in Neonate after 5 hours	
X18	iPT	Intensive Phototherapy done on Neonates	
X19	fSBR_miJ	Serum Bilirubin levels falling after having mild Jaundice	

X20	rSBR_miJ	Serum Bilirubin levels rising after having mild Jaundice
X21	fSBR_moJ	Serum Bilirubin levels falling after having moderate Jaundice
X22	rSBR_moJ	Serum Bilirubin levels rising after having moderate Jaundice
X23	cSBR_4h	Check Serum Bilirubin levels in Neonate after 4 hours
X24	inform_NICU	Inform Neonatal Intensive care unit
X25	fSBR_sJ	Serum Bilirubin levels falling after having severe Jaundice
X26	rSBR_sJ	Serum Bilirubin levels rising after having severe Jaundice
X27	kus _n	Keep Neonate under supervision
X28	dPT	Double Phototherapy done on neonates

X29	EXTR	Neonate undergoes exchange transfusion
X30	dSBR2_6h	Determine Serum Bilirubin levels in Neonate again after 6 hours
X31	baPHI	Bilirubin above Phototherapy line
X32	cSBR2_5h	Check Serum Bilirubin levels in Neonate again after 5 hours
X33	rSBR2_moJ	Serum Bilirubin levels rising again after having moderate Jaundice
X34	b50bPHI	Bilirubin 50 mg/l below Phototherapy line
X35	ep	End Protocol
X36	nb _{HCP}	HCP mental model state for registering that Neonate is Born

X37	vsy_{HCP}	HCP mental model state for registering that neonates skin is visually yellow
X38	ps_{HCP}	HCP mental model state for registering that Neonate has pale stools
X39	du_{HCP}	HCP mental model state for registering that Neonate has dark urine
X40	$mbgi_{HCP}$	HCP mental model state for registering that Neonate has incompatibility with maternal blood type
X41	$orfnh_{HCP}$	HCP mental model state for realising and observing the risk factors for neonatal hyperbilirubinemia
X42	auc_{HCP}	HCP mental model state for assessing the underlying causes for neonatal hyperbilirubinemia, needed for future purposes if initial treatments do not work.
X43	bf_{HCP}	HCP mental model state to initiate Neonate with

Base Level

-

		breastfeeding procedure within 1 hour of birth	HCP Mental Model
X44	dSBR _{HCP}	HCP mental model state to Determine Serum Bilirubin concentration of neonate in the blood by taking a blood sample	
X45	miJ _{HCP}	HCP mental model state to register that Neonate has mild Jaundice	
X46	moJ _{HCP}	HCP mental model state to register that Neonate has moderate Jaundice	
X47	sJ _{HCP}	HCP mental model state to register that Neonate has severe Jaundice	
X48	tmc _{HCP}	HCP mental model state to affectuate transferring Neonate to maternal care	
X49	sPT _{HCP}	HCP mental model state to administer Neonate with Single Phototherapy	
X50	aNICU _{HCP}	HCP mental model state to administer Neonate to the	

Neonatal intensive care unit		
X51	cSBR_6h _{HCP}	HCP mental model state to affectuate checking Serum Bilirubin levels after 6 hours
X52	cSBR_5h _{HCP}	HCP mental model state to affectuate checking Serum Bilirubin levels after 5 hours
X53	iPT _{HCP}	HCP mental model state to administer Neonate with Intensive Phototherapy
X54	fSBR_miJ _{HCP}	HCP mental model state to register falling Serum Bilirubin levels in Neonate after having mild Jaundice
X55	rSBR_miJ _{HCP}	HCP mental model state to register rising Serum Bilirubin levels in Neonate after having mild Jaundice
X56	fSBR_moJ _{HCP}	HCP mental model state to register falling Serum Bilirubin levels in Neonate after having moderate Jaundice

X57	rSBR_moJ _{HCP}	HCP mental model state to register rising Serum Bilirubin levels in Neonate after having moderate Jaundice
X58	cSBR_4h _{HCP}	HCP mental model state to affectuate checking Serum Bilirubin levels after 4 hours
X59	inform_NICU _{HCP}	HCP mental model state to affectuate informing Neonatal Intensive Care Unit
X60	fSBR_sJ _{HCP}	HCP mental model state to register falling Serum Bilirubin levels in Neonate after having severe Jaundice
X61	rSBR_sJ _{HCP}	HCP mental model state to register rising Serum Bilirubin levels in Neonate after having severe Jaundice
X62	ku _{HCP}	HCP mental model state to affectuate keeping Neonate under supervision

X63	dPT _{HCP}	HCP mental model state to administer Neonate with Double Phototherapy
X64	EXTR _{HCP}	HCP mental model state to administer Neonate with exchange transfusion
X65	dSBR2_6h _{HCP}	HCP mental model state to Determine Serum Bilirubin levels in Neonate again after 6 hours
X66	baPHI _{HCP}	HCP mental model state to register that neonate's Bilirubin levels are above Phototherapy line
X67	cSBR2_5h _{HCP}	HCP mental model state to Check Serum Bilirubin levels in Neonate again after 5 hours
X68	rSBR2_moJ _{HCP}	HCP mental model state to register Serum Bilirubin levels rising again after having moderate Jaundice
X69	b50bPHI _{HCP}	HCP mental model state to register that neonate's Bilirubin levels 50 mg/l below Phototherapy line

X70	ep_{HCP}	HCP mental model state to affectuate Ending the Protocol	
X71	$W_{vsy_{HCP}, orfnh_{HCP}}$	Reified representation state for connection weight $\omega_{vsy_{HCP}, orfnh_{HCP}}$	
X72	$W_{fSBR_sJ_{HCP}, dPT_{HCP}}$	Reified representation state for connection weight $\omega_{fSBR_sJ_{HCP}, dPT_{HCP}}$	First
X73	$W_{dPT_{HCP}, cSBR2_5h_{HCP}}$	Reified representation state for connection weight $\omega_{dPT_{HCP}, cSBR2_5h_{HCP}}$	Reification Level
X74	$W_{fSBR_moJ_{HCP}, tmc_{HCP}}$	Reified representation state for connection weight $\omega_{fSBR_moJ_{HCP}, tmc_{HCP}}$	
X75	$W_{fSBR_miJ_{HCP}, kus_{HCP}}$	Reified representation state for connection weight $\omega_{fSBR_miJ_{HCP}, kus_{HCP}}$	

X76	nb_{AIC}	AIC mental model state for registering that Neonate is Born
X77	vsy_{AIC}	AIC mental model state for registering that Neonate Skin is Visually Yellow
X78	ps_{AIC}	AIC mental model state for registering that Neonate has pale stools
X79	du_{AIC}	AIC mental model state for registering that Neonate has dark urine
X80	$mbgi_{AIC}$	AIC mental model state for registering that Neonate has incompatibility with maternal blood type
X81	$orfnh_{AIC}$	AIC mental model state for Observing risk factors in

		neonate for neonatal hyperbilirubinemia	
X82	auc_{AIC}	AIC mental model state for Assessing underlying causes for neonatal hyperbilirubinemia, needed for future purposes if initial treatments do not work.	
X83	bf_{AIC}	AIC mental model state for monitoring that breastfeeding is done within 1 hour of birth	Base Level - AIC Mental Model
X84	$dSBR_{AIC}$	AIC mental model state for monitoring that serum bilirubin concentration of neonate in the blood is determined by taking a blood sample	
X85	miJ_{AIC}	AIC mental model state for registering that Neonate has Mild Jaundice	
X86	moJ_{AIC}	AIC mental model state for registering that Neonate has Moderate Jaundice	

X87	sJ_{AIC}	AIC mental model state for registering that Neonate Severe Jaundice
X88	tmc_{AIC}	AIC mental model state for monitoring that Neonate has to be transferred to maternal care
X89	sPT_{AIC}	AIC mental model state for monitoring that Single Phototherapy has to be done on Neonate
X90	$aNICU_{AIC}$	AIC mental model state for monitoring a Neonates admission to the Neonatal Intensive care unit
X91	$cSBR_6h_{AIC}$	AIC mental model state for monitoring Serum Bilirubin levels check done to Neonate after 6 hours
X92	$cSBR_5h_{AIC}$	AIC mental model state for monitoring Serum

		Bilirubin levels check done to Neonate after 5 hours
X93	iPT_{AIC}	AIC mental model state for monitoring that Intensive Phototherapy is done on Neonates
X94	$fSBR_miJ_{AIC}$	AIC mental model state for monitoring falling Serum Bilirubin levels after having mild Jaundice
X95	$rSBR_miJ_{AIC}$	AIC mental model state for monitoring rising Serum Bilirubin levels after having mild Jaundice
X96	$fSBR_moJ_{AIC}$	AIC mental model state for monitoring falling Serum Bilirubin levels after having moderate Jaundice
X97	$rSBR_moJ_{AIC}$	AIC mental model state for monitoring rising Serum Bilirubin levels after having moderate Jaundice
X98	$cSBR_4h_{AIC}$	

		AIC mental model state for monitoring a Serum Bilirubin levels done to Neonate after 4 hours
X99	inform_NCU _{AIC}	AIC mental model state to monitor that the Neonatal Intensive Care Unit is informed of a neonates worsening condition
X100	fSBR_sJ _{AIC}	AIC mental model state to register falling Serum Bilirubin levels in Neonate after having severe Jaundice
X101	rSBR_sJ _{AIC}	AIC mental model state to register rising Serum Bilirubin levels in Neonate after having severe Jaundice
X102	kus _{AIC}	AIC mental model state to monitor that a Neonate is kept under supervision

X103	dPT _{AIC}	AIC mental model state to monitor that Double Phototherapy is done to neonates
X104	EXTR _{AIC}	AIC mental model state to monitor that exchange transfusion is carried out on a neonate
X105	dSBR2_6h _{AIC}	AIC mental model state to monitor the determining of Serum Bilirubin levels in Neonate again after 6 hours
X106	baPHI _{AIC}	AIC mental model state to register that neonate's Bilirubin levels are above Phototherapy line
X107	cSBR2_5h _{AIC}	AIC mental model state to monitor that the Serum Bilirubin levels in Neonate are checked again after 5 hours
X108	rSBR2_moJ _{AIC}	

		AIC mental model state to register Serum Bilirubin levels rising again after having moderate Jaundice	
X109	$b50bPHI_{AIC}$	AIC mental model state to register that neonate's Bilirubin levels 50 mg/l below Phototherapy line	
X110	ep_{AIC}	AIC mental model state for monitoring that the Protocol is ended.	
X111	$MW_{vsy_{HCP}, orfnh_{HCP}}$	Reified representation state for persistence factor $\mu W_{vsy_{HCP}, orfnh_{HCP}}$ for reified representation state $W_{vsy_{HCP}, orfnh_{HCP}}$	
X112	$MW_{fSBR_sJ_{HCP}, dPT_{HCP}}$	Reified representation state for persistence factor $\mu W_{fSBR_sJ_{HCP}, dPT_{HCP}}$ for reified representation state $W_{fSBR_sJ_{HCP}, dPT_{HCP}}$	Second Reification Level

X113	$\mathbf{Mw}_{dPT_{HCP,cSBR2_5h_{HC}}}$ P	Reified representation state for persistence factor $\mu \mathbf{W}_{dPT_{HCP,cSBR2_5h_{HCP}}}$ for reified representation state $\mathbf{W}_{dPT_{HCP,cSBR2_5h_{HCP}}}$
X114	$\mathbf{Mw}_{fSBR_moJ_{HCP,tmc_{HC}}}$ P	Reified representation state for persistence factor $\mu \mathbf{W}_{fSBR_moJ_{HCP,tmc_{HC}}}$ P for reified representation state $\mathbf{W}_{fSBR_moJ_{HCP,tmc_{HCP}}}$
X115	$\mathbf{Mw}_{fSBR_miJ_{HCP,kus_{HCP}}}$	Reified representation state for persistence factor $\mu \mathbf{W}_{fSBR_miJ_{HCP,kus_{HCP}}}$ for reified representation state $\mathbf{W}_{fSBR_miJ_{HCP,kus_{HCP}}}$
X116	$cf_{v_{sy}}$	Context factor state v_{sy_n}
X117	cf_{ps}	Context factor state ps_n

X118	cf_{du}	Context factor state du_n	
X119	cf_{mbgi}	Context factor state $mbgi_n$	
X120	cf_{miJ}	Context factor state miJ	
X121	cf_{moJ}	Context factor state moJ	Base Level
X122	cf_{sJ}	Context factor state sJ	- Context Factor
X123	cf_{fSBR_miJ}	Context factor state $fSBR_miJ$	States
X124	cf_{rSBR_miJ}	Context factor state $rSBR_miJ$	
X125	cf_{fSBR_moJ}	Context factor state $fSBR_moJ$	
X126	cf_{rSBR_moJ}	Context factor state $rSBR_moJ$	
X127	cf_{fSBR_sJ}	Context factor state $fSBR_sJ$	
X128	cf_{rSBR_sJ}	Context factor state $rSBR_sJ$	

X129	cf_{rSBR2_moJ}	Context factor state $rSBR2_moJ$	
X130	cf_{baPHI}	Context factor state $baPHI$	
X131	$cf_{b50bPHI}$	Context factor state $b50bPHI$	
X132	$monitor_{orfnh}$	Monitor state for AIC mental model for monitoring world state $orfnh$	
X133	$monitor_{tmc}$	Monitor state for AIC mental model for monitoring world state $orfnh$	
X134	$monitor_{kus}$	Monitor state for AIC mental model for monitoring world state $orfnh$	First Reification Level
X135	$monitor_{dPT}$	Monitor state for AIC mental model for monitoring world state $orfnh$	

X136	monitor _{cSBR2_5h}	Monitor state for AIC mental model for monitoring world state orfnh
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It is important to note that, particularly in Scenario 3 where the AI Coach is present in the simulations, the AI Coach states occupy the same space and state numbers, namely from X76 till X110 for both neonatal medical protocols. This signifies the AI Coach is adaptable and flexible to recognise the symptoms for either of the neonatal medical protocols and choose the path corresponding to those symptoms and thus on the pathway of the correct medical guidelines, if any of them are stored within its mental model. It also shows that the AIC remains the same as choosing a different pathway in its mental model corresponding to a different medical guideline is simply a subsection of the AIC mental model, and that even more medical protocols could be stored in the same AIC mental model without changing any dynamics of how the AIC interacts with the HCP and inducing adherence to guidelines.

In the attached Appendix, tables for the role matrices that are used for the simulation of the scenarios can be found.

5 Results

This section depicts the final output of the simulation charts modelling the self-adaptive networks for the two medical protocols in their corresponding three scenarios.

5.1 Simulation Outputs for the Neonatal Hypoglycemia Medical Protocol

Figure 5 shows Scenario 1 of the Neonatal Hypoglycemia treatment where the HCP performs the treatment and follows the corresponding protocol with a perfect mental model without any flaws. It can be observed that each real-world state in the protocol as well as the mental model states are executed rapidly in succession without any steps taken being delayed in their execution (see scenario 2) or not being executed properly and reaches a level between 0.8 to 1 signifying a high degree of completion. In addition to the protocol being followed properly the learning of HCP can be seen to be rising to full saturation as the treatment progresses. However, note that the W-state lines, starting at the 0.9 threshold, have a slight initial decrease in value before recovering and increasing to a full threshold as the treatment progresses and the HCP completes the corresponding protocol (vulnerable) steps in the real world which boosts learning. Note here that a W-state which corresponds to the mental model state which has to be

followed later on in the protocol has a larger decrease before recovering when compared to a W-state for which the corresponding mental model state has to be followed earlier on in the protocol. The MW-states beginning from 0.8, however, do not decrease in value but only increase to full saturation as the treatment progresses, albeit in a similar fashion to the trend observed while observing the W-states, where MW-states corresponding to mental model states which are followed later on in the protocol increase much more slower when compared to MW-states for which the corresponding mental model states are followed earlier on in the protocol. The entire protocol is completed by $t=25$.

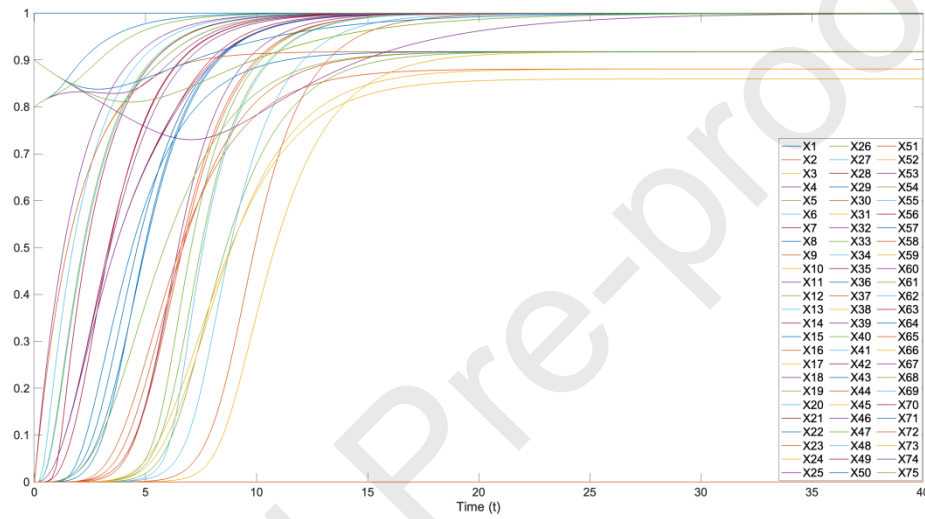


Fig. 5 - Simulation Results for Neonatal Hypoglycemia Treatment Scenario 1

Figure 6 shows Scenario 2 of the Neonatal Hypoglycemia treatment where the HCP follows the corresponding protocol with an imperfect mental model having flaws in the connections between mental model states vulnerable to encompass mistakes. It can be witnessed that the W- and M- states, while following the same trend of an initial decline followed by a recovery, have a very gradual and late rise, and as before, also have a slower rise for the W-states for which the mental model states appear later on in the protocol. The corresponding real-world and mental model states to the W-states also follow a late and gradual rise. As an effect of these steps not being completely executed and incurring delays in reaching their required threshold levels, firstly the mental model states which are not vulnerable as well as their corresponding world states, that are executed before the vulnerable states in the protocol rise rapidly and reach their final thresholds. However, states executed after the vulnerable states, which have not been followed properly, do not reach their final thresholds rapidly. Instead they rise to a certain threshold and stay fairly stagnant until the vulnerable mental model states, corresponding W- and M- states and world states rise rapidly after a delay, signifying that the HCP has realized the flaw and corrects the mistake to complete the step properly. In case of this model, around $t=100$ is where W-states and their corresponding vulnerable mental model and world states rise rapidly to their final thresholds and it is here where states executed after the vulnerable mental model states which were

stagnant between 0.8 and 0.9 further increase to level 1 signifying their correct implementation after the previous flaw has been corrected. The entire protocol is completed by $t=200$.

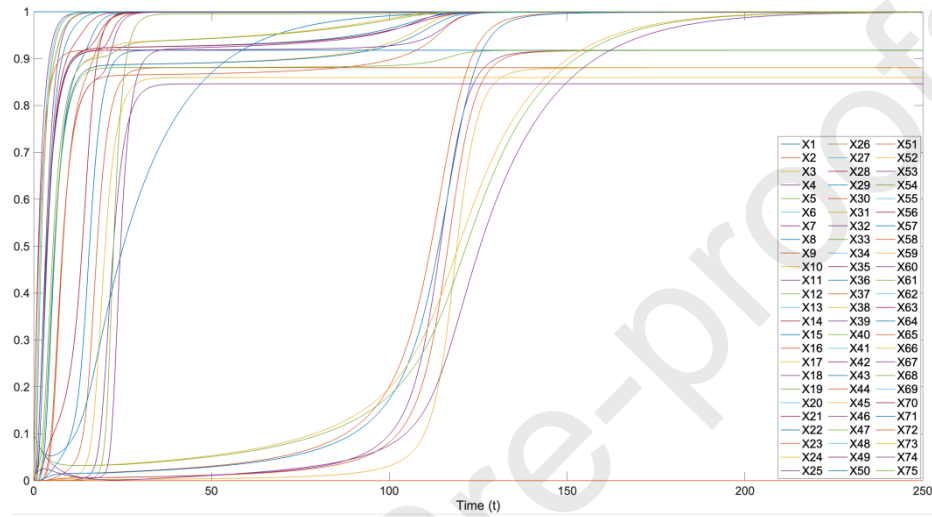


Fig. 6 - Simulation Results for Neonatal Hypoglycemia Treatment Scenario 2

Figure 7a and 7b shows Scenario 3a of the Neonatal Hypoglycemia treatment where the HCP performs the treatment and follows the corresponding protocol with a perfect mental model without any flaws with the additional presence of an AI Coach. It can be observed that the model follows the exact trend as scenario 1 of the Neonatal Hypoglycemia treatment. Specifically, it can be seen that the real-world state in the protocol as well as the mental model states are executed rapidly in succession without any steps taken being delayed in their execution or not being executed properly and reaches a threshold level between 0.8 to 1 signifying a high degree of completion. In addition to the protocol being followed properly the learning of HCP, can be seen to be rising to full saturation as the treatment progresses. In this case, possibly due to the additional presence of the AIC, there is no initial decrease in the threshold levels of the W-states. The MW-states beginning from 0.8, also, do not decrease in value but only increase to full saturation as the treatment progresses, albeit in a similar fashion to the trend observed while observing the W-states, where MW-states corresponding to mental model states which are followed later on in the protocol increase afterwards when compared to MW-states for which the corresponding mental model states are followed earlier on in the protocol. The entire protocol is completed by $t=25$. It can be further noted that unlike in scenario 2, there is not any state which increases to a certain threshold level and stays stagnant and then further increases to a final threshold level, due to other mental model states and W-states having a slow and gradual rise. Upon closer inspection of the model in figure 7b, it can also be seen that the monitor states stay dormant due to the dynamic nature of the AIC which does not trigger support signals if the doctor does not commit a mistake and if the protocol is being followed to a high degree of satisfaction.

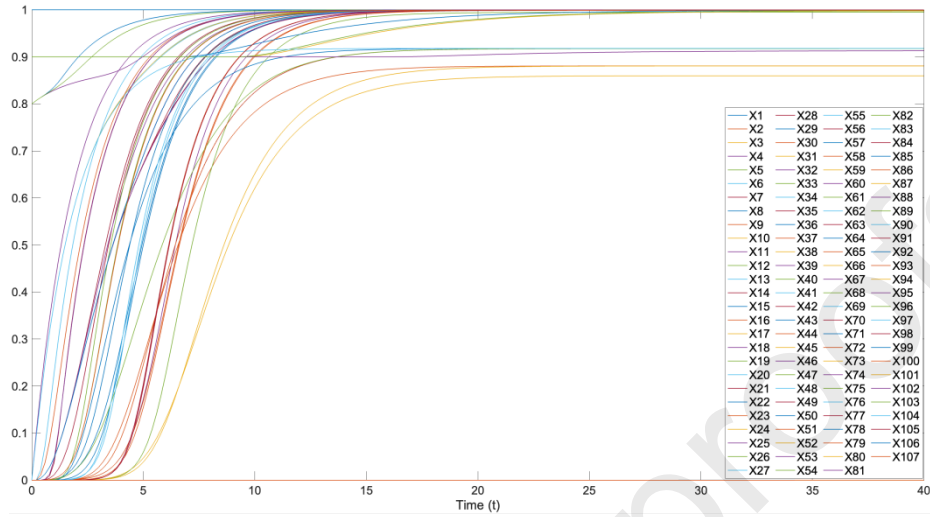
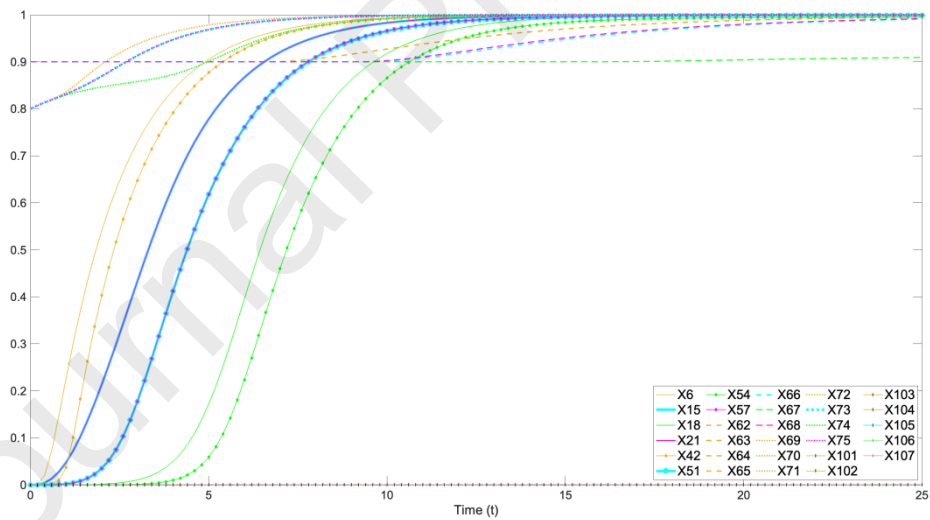


Fig. 7a - Simulation Results for Neonatal Hypoglycemia Treatment Scenario 3a



LEGEND

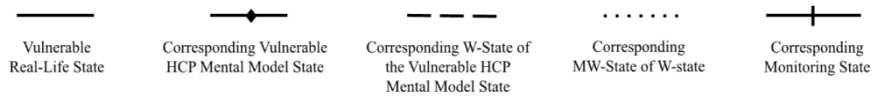


Figure 7b - Magnified Chart of the Monitor and Adaptive States in the Reification Levels and Vulnerable HCP Mental Model and Corresponding Real-World States in the Base Level of the Neonatal Hypoglycemia Treatment: Scenario 3a

Figure 8a and 8b below, depict scenario 3b of the Neonatal Hypoglycemia treatment where the HCP follows the corresponding protocol with an imperfect mental model having flaws in the connections between mental model states vulnerable to mistakes, however there is an added presence of the AI coach which serves to correct those deviations. It can be observed in figure 8b that unlike in scenario 2, while there is an initial staggered execution of the vulnerable HCP mental model states and corresponding real-world and W- and M- states, this deviation is quickly fixed the moment there is a sudden and steep rise in the monitoring states around $t=5$, which occurs few moments after the protocol has started as the AIC recognizes mistakes committed through incomplete and faulty execution of the protocol as seen in the real-world states by the AIC which then further relays support signals to the mental model of the HCP. When this happens, there is a sudden and rapid rise firstly in persistence, then in the learning in W-states, thereafter a further increase in the correct execution of the mental model state, solidifying the correction of the mistake in the HCP mental model and lastly this translates into an increase in the threshold levels of the corresponding real-world states. The correction of the deviation in the protocol thus happens far quicker than when compared to scenario 2 where the correction happened around $t=100$, as here the entire protocol is completed by $t=50$. This also means that unlike in scenario 2, all real-world states rise rapidly to their final thresholds, and even states executed after the vulnerable states, which have not been initially followed properly, do not experience any stagnant plateau, but only just have a slight drop in the steepness of the rise before quickly continuing to rise to their final thresholds.

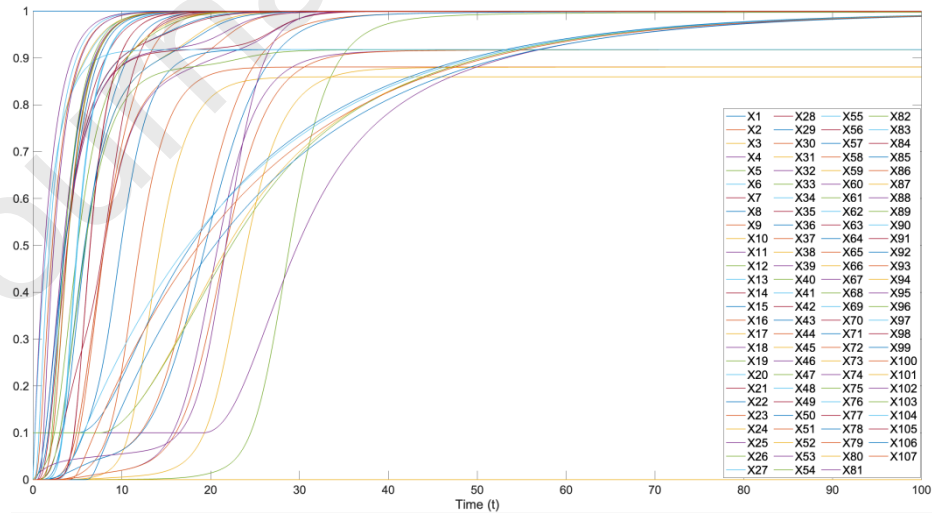
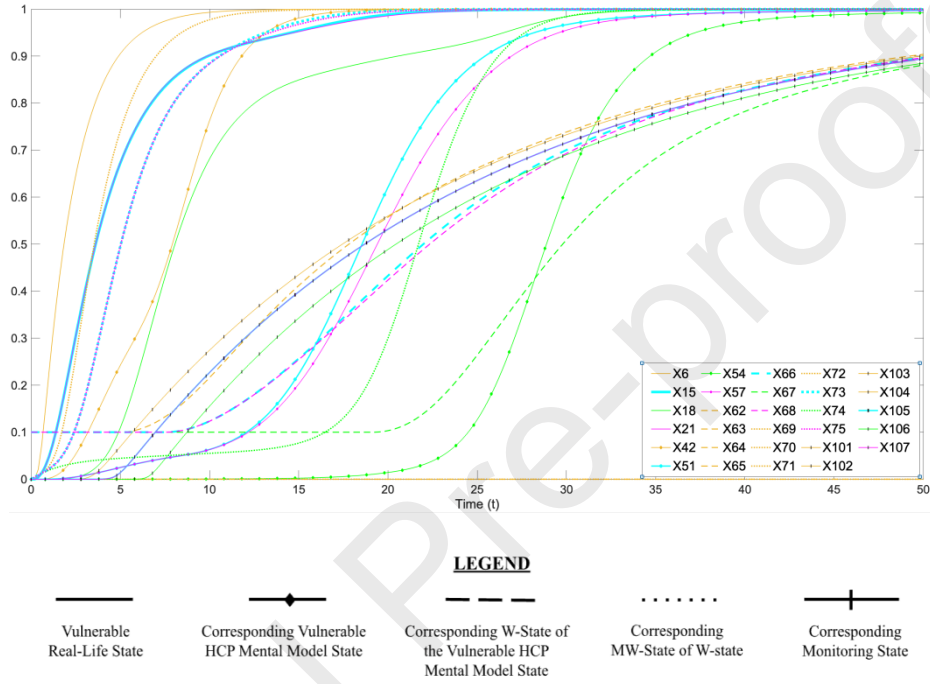


Figure 8a - Simulation Results for Neonatal Hypoglycemia Treatment: Scenario 3b**Figure 8b** - Magnified Chart of the Monitor and Adaptive States in the Reification Levels and Vulnerable HCP Mental Model and Corresponding Real-World States in the Base Level of the Neonatal Hypoglycemia Treatment: Scenario 3b

5.2 Simulation Outputs for the Neonatal Hyperbilirubinemia Medical Protocol

Figure 9 shows scenario 1 of the Neonatal Hyperbilirubinemia treatment where the HCP performs the treatment and follows the corresponding protocol with a perfect mental model without any flaws. It can be extrapolated upon close inspection that the states in the observed model follow the exact same pattern and trends as of those in scenario 1 of the Neonatal Hypoglycemia treatment. Each real-world state in the protocol as well as the mental model states are executed rapidly in succession without any steps taken being delayed in their execution or not being executed properly and reaches a level between 0.8 to 1 signifying a high degree of completion. The learning of HCP can be seen to be rising to full saturation as the treatment progresses with the

W-states, starting at the 0.9 threshold, have a slight initial decrease in value before recovering and increasing to a full threshold as the treatment progresses. As in scenario 1 of the Neonatal Hypoglycemia treatment, the W-states which are executed later on in the protocol have a larger initial decrease when compared to a W-state for which the corresponding mental model state has to be followed earlier on in the protocol. The MW-states beginning from 0.8, however, do not decrease in value but only increase to a full threshold level as the treatment progresses, however with MW-states followed later on in the protocol increasing more gradually when compared to MW-states which are followed earlier on in the protocol. Protocol is completed by $t=25$.

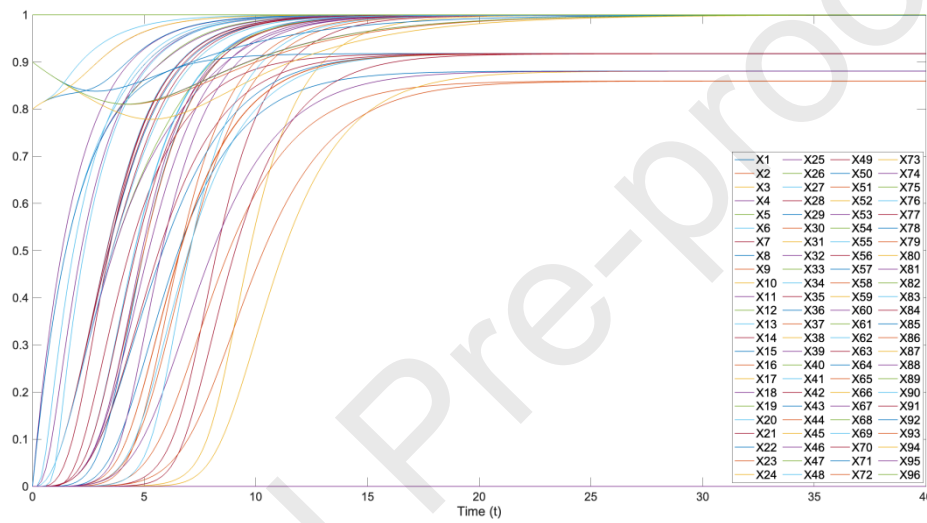


Fig. 9 - Simulation Results for Neonatal Hyperbilirubinemia Protocol Scenario 1

Fig. 10 shows scenario 2 of the Neonatal Hyperbilirubinemia treatment where the HCP follows the corresponding protocol with an imperfect mental model having flaws in the connections between mental model states vulnerable to mistakes. Keeping in line with the same trends as seen in scenario 2 of the Neonatal Hypoglycemia treatment, the W- and M- states, along with their corresponding mental model states and real-world states have a very gradual and late rise, and as before, also have a slower rise for the W-states for which the mental model states appear later on in the protocol. Further on, as described before an effect of these steps not being completely executed and incurring delays in reaching their required threshold levels leads to states executed after the vulnerable states, rise to a certain threshold and stay fairly stagnant until the vulnerable mental model states, corresponding W- and M- states and world states rise rapidly after a delay, signifying that the HCP has realized the flaw and corrects the mistake to complete the step properly. In case of this model, at $t=100$ and then again at $t=150$ is where W-states start rising rapidly to their final thresholds. The moment the rapid rise of the mental model and W- and M- states starts occurring, the states executed after the vulnerable mental model states which were stagnant between 0.8 and 0.9 finally

increase to their final threshold signifying their correct implementation after the previous flaw has been corrected. The entire protocol is completed by $t=250$.

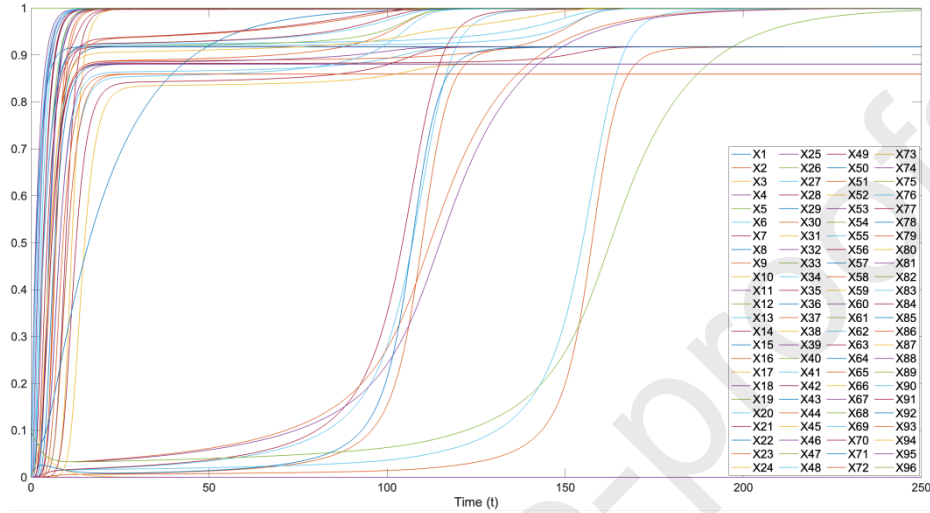


Fig. 10 - Simulation Results for Neonatal Hyperbilirubinemia Protocol Scenario 2

Figure 11a and 11b shows Scenario 3a of the Neonatal Hyperbilirubinemia treatment where the HCP performs the treatment and follows the corresponding protocol with a perfect mental model without any flaws with the additional presence of an AI Coach. It can be observed that the model follows the exact trend as scenario 1 of the Neonatal Hyperbilirubinemia treatment. Specifically, it can be seen that the real-world state in the protocol as well as the mental model states are executed rapidly in succession without any steps taken being delayed in their execution or not being executed properly and reaches a threshold level between 0.8 to 1 signifying a high degree of completion. In addition to the protocol being followed properly the learning of HCP, can be seen to be rising to full saturation as the treatment progresses. The MW-states beginning from 0.8, also, do not decrease in value but only increase to full saturation as the treatment progresses, albeit in a similar fashion to the trend observed while observing the W-states, where MW-states corresponding to mental model states which are followed later on in the protocol increase afterwards when compared to MW-states for which the corresponding mental model states are followed earlier on in the protocol. The entire protocol is completed by $t=25$, just as in scenario 1. It can be further noted that unlike in scenario 2, there is not any state which increases to a certain threshold level and stays stagnant and then further increases to a final threshold level, due to other mental model states and W-states having a slow and gradual rise.

Upon closer inspection of the model in figure 11b, it can also be seen that the monitor states stay dormant due to the dynamic nature of the AIC which does not trigger support signals if the doctor does not commit a mistake and if the protocol is being followed to a high degree of satisfaction. This not only displays the dynamic nature of the AIC with respect to taking into account different scenarios of the protocol being followed but also with respect to recognizing and functioning in different environments when it comes to following different medical protocols.

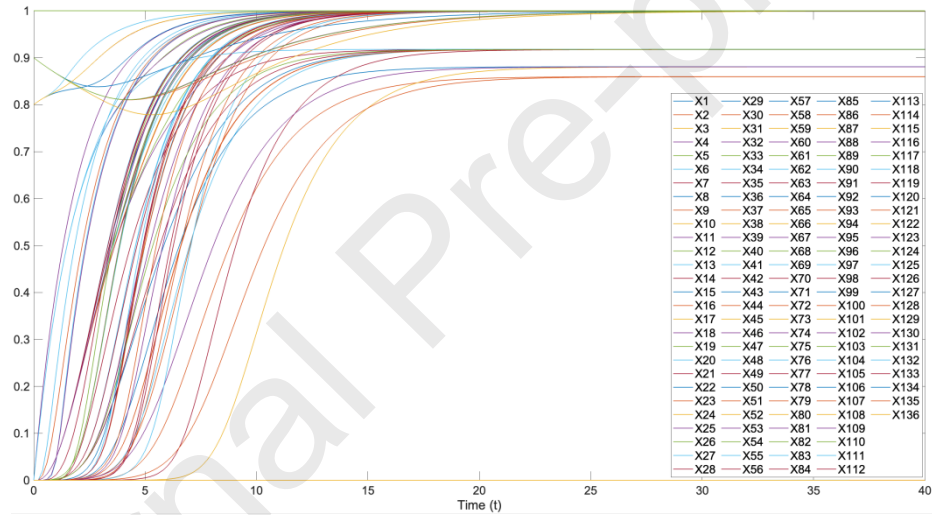


Fig. 11a - Simulation Results for Neonatal Hyperbilirubinemia Protocol Scenario 3a

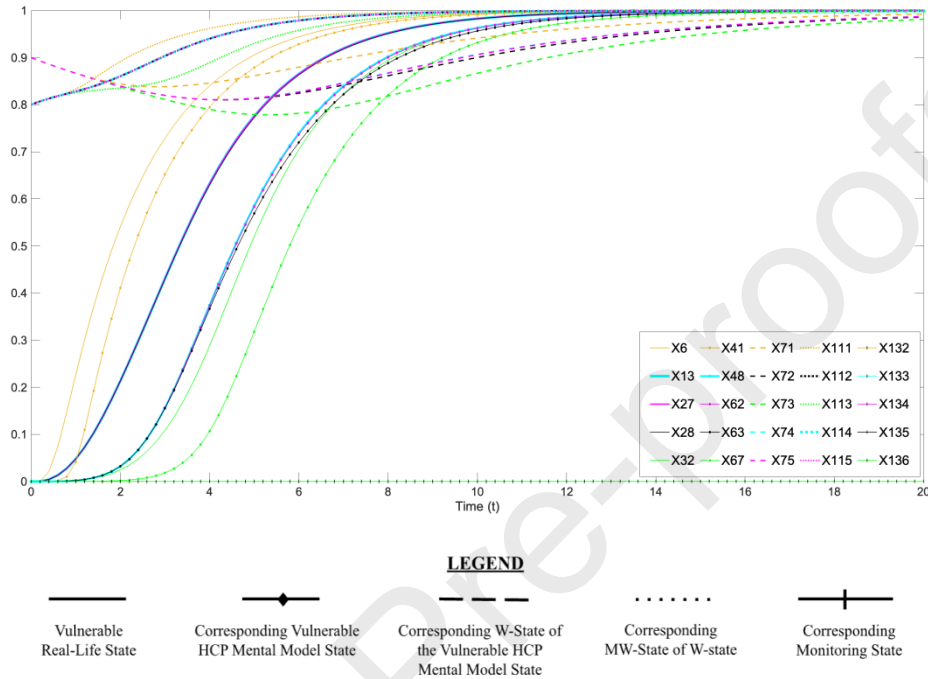


Figure 11b - Magnified Chart of the Monitor and Adaptive States in the Reification Levels and Vulnerable HCP Mental Model and Corresponding Real-World States in the Base Level of the Neonatal Hyperbilirubinemia Treatment: Scenario 3a

Figure 12a and 12b below, depict scenario 3b of the Neonatal Hyperbilirubinemia treatment where the HCP follows the corresponding protocol with an imperfect mental model having flaws in the connections between mental model states vulnerable to mistakes, however there is an added presence of the AI coach which serves to correct those deviations. It can be observed that figure 12a follows a very similar patterns and overall trend of the states when compared to scenario 3b of the Neonatal Hypoglycemia protocol. This emphasizes the dynamic nature of the model irrespective of various different treatments. In figure 12b, unlike in scenario 2, while there is an initial staggered execution of the vulnerable HCP mental model states and corresponding real-world and W- and M- states, (including an initial drop in the W-state levels which start from 0.1), this deviation is quickly fixed with the sudden and steep rise in the monitoring states around $t=5$, which occurs few moments after the protocol has started, since the AIC recognizes mistakes committed through incomplete and faulty execution of the protocol as seen in the real-world and then further relays support signals to the mental model of the HCP. When this happens, there is a sudden and rapid rise firstly in persistence, then in the learning in W-states, thereafter a further increase in the correct

execution of the mental model state, solidifying the correction of the mistake in the HCP mental model and lastly this translates into an increase in the threshold levels of the corresponding real-world states. The correction of the deviation in the protocol thus happens far quicker than when compared to scenario 2 where the correction happened between $t=100$ and $t=150$, as here the entire protocol is completed by $t=60$. This also means that unlike in scenario 2, all real-world states rise rapidly to their final thresholds, and even states executed after the vulnerable states, which have not been initially followed properly, do not experience any stagnant plateau, but only just have a slight drop in the steepness of the rise before quickly continuing to rise to their final thresholds.

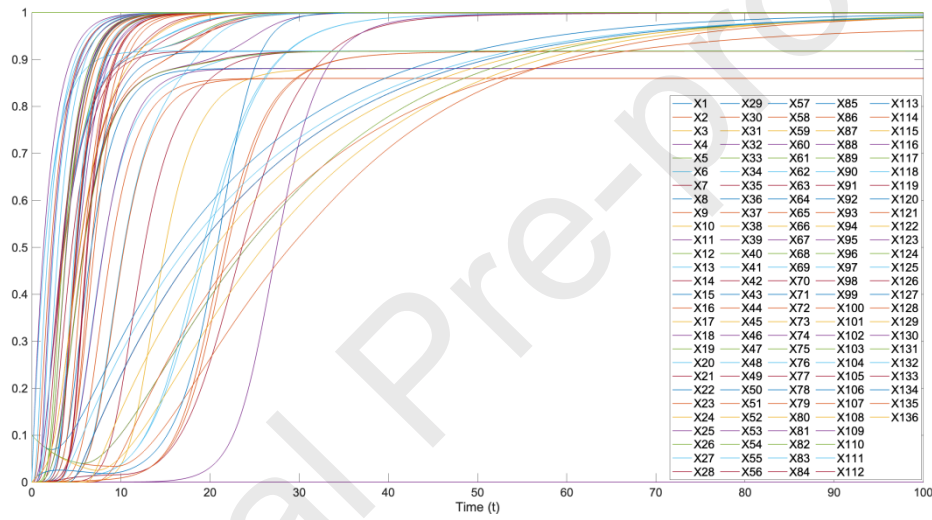


Figure 12a - Simulation Results for Neonatal Hyperbilirubinemia Treatment: Scenario 3b

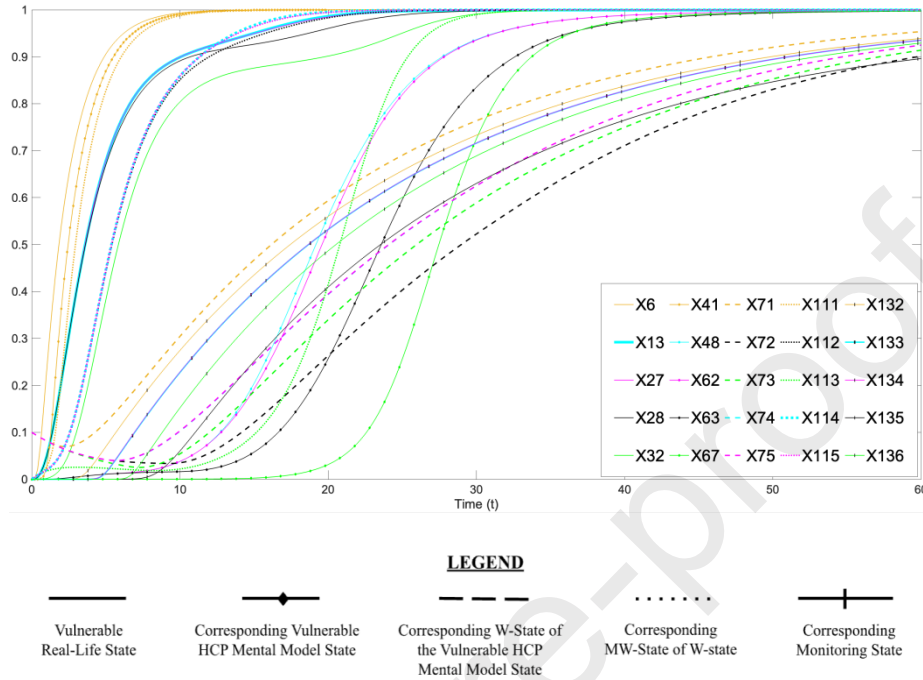


Figure 12b - Magnified Chart of the Monitor and Adaptive States in the Reification Levels and Vulnerable HCP Mental Model and Corresponding Real-World States in the Base Level of the Neonatal Hyperbilirubinemia Treatment: Scenario 3b

5.3 Results Analysis

The above simulations show a clear common trend the AIC plays in promoting the adherence of medical health care practitioner to guidelines. Upon careful inspection of the traces in the dynamic model, it can be seen that in Scenario 1 for both simulations, the state levels which correspond to the full execution of the states and following them, increase rapidly to a certain levels and then stay stagnant, meaning completed, while the 2nd and 1st order adaptive states keep increasing, thus persistence and learning increases after a small stagnation in growth. However, with Scenario 2, a significant delay in the increase of the vulnerability states and the corresponding **W**- and **M**-states is witnessed while the other states increase to a certain level before stagnating and having to wait for the states which have incurred deviations to increase so they can further keep increasing to their required thresholds. For instance, while in Hyperbilirubinemia Scenario 1, states took 60 time units to increase fully, in Scenario 2 it took more than 250 time units. In the real world, this would play out as the HCP forgetting to perform a certain action in the protocol, or not perform it to a satisfactory level initially, and then move on to the next steps before realising something more needs to be done for the step where the mistake was made and going back to that step with significant delay and correcting it while the other states remain stagnant until the mistake is corrected. As can be seen here, persistence **M**-states traces, and **W**-state traces are very low and have a very slow rate of increase. In Scenario 3 however, with

the intervention of the AIC, it can be observed that the depictions of the traces almost identically match those in Scenario 1, with the monitoring states suppressed without triggering any actions as the mental model is not flawed and the vulnerable HCP mental model states and the corresponding real-world state in the protocol reaching full saturation or their execution completion very rapidly and this knowledge of the presence of the real world states being transferred to the AIC which does not intervene. With scenario 3b, however, the AIC detects an error in the process of the protocol being followed in the real-world states and thus intervenes by sending a signal to the monitor states to become active and help improve the HCP mental model flaws and subsequently the HCP's knowledge. The sudden and steep rise of the monitor states after being dormant till the AIC recognises a flaw and sends a corresponding signal to the monitor state, further leads to a sudden rapid and sharp rise in persistence and learning of the W-states of the vulnerable HCP mental model states which were flawed and caused an error while following the protocol in the real world. This further causes a fast correction of the previously seen delay in scenario 2 for the doctor to realise the error committed following the protocol and go back to ensure its correct execution which then further enables other steps in the protocol to also be fully completed after being stagnated at a certain level of completion for the duration of the delay. In fact, the charts of scenario 3b are visually the horizontally contracted version of the charts seen in scenario 2 except the presence of monitoring states and the AIC mental model states. Thus, in scenario 3b in both medical treatments, the vulnerable states reach full saturation by around 40 time units while in scenario 2 of both medical treatments takes up to 150 units to reach full saturation.

6 Discussion

This article presented an adaptive second-order agent network model that covers three scenarios in a medical setup, where health care practitioners are following medical protocols either perfectly or imperfectly with the support of an AI Coach. These models are developed via an adaptive network-oriented modeling approach and simulated through a specialized software environment in MATLAB. The models depict the representation and processing of the actors' mental simulation, their mental model adaptability, interactions with the world, and the dynamics of these interactions. Through integrating higher-order adaptive networks, a dynamic and evolving system that caters to the changing demands and nuances of the medical profession is created and in this way this article addresses understanding and enforcing medical protocol adherence. Such network-oriented modeling techniques have also been applied in different domains such as addressing communication with parents and postpartum depression (Weigl, Jabeen, Treur, Taal, Roelofsma, 2023; 2024) and addressing respiratory problems of new-born babies (Mokadem, Jabeen, Treur, Taal, Roelofsma, 2024a; 2024b; Sebahi, Jabeen, Treur, Taal, Roelofsma, 2024a; 2024b).

In conclusion, the model successfully demonstrates how the incorporation of an AIC can effectively and dynamically intervene to prevent deviations from various different medical protocols happening in the real-world at the hands of a HCP. The model further illustrates various learning effects, including internal Hebbian learning, showcasing their influence on mental model structures. Simulation experiments highlight the role of AI-Coach in fostering adherence to guidelines as the AI-Coach

monitors team actions, issuing alerts to the health care practitioners upon identifying potential mistakes.

The simulation experiments highlight how incomplete knowledge, and a flawed mental model could have severe implications in the proper execution of a protocol in the real world. They also underscore the role of an AI-coach in fostering adherence to guidelines as the AI-coach monitors the implementation of the protocol and the actions of the HCP, issuing alerts to the health care practitioners upon identifying potential mistakes. Simulation when a part of the protocol is not fully executed in the real world which in turn provides knowledge reinforcement and support. The simulation results further underscore the effects of this intervention, such as the prevention of delay in a state's full execution and increase in persistence and learning when a mistake is committed with the intervention of the AIC and the effective dynamic safeguard provided by the monitor states which act in accordance with the contexts of the different scenario's.

This research is an invaluable contribution to optimizing adherence to neonatal treatment guidelines, thereby fostering a safer, more efficient neonatal care environment. The presented model has a few limitations, namely that this is a theoretical model which does not have any practical experience. It also assumes the AI Coach monitors the medical process perfectly and also communicates with the HCP perfectly. However, in the real-world effective mechanisms to conduct the AIC's communication and monitoring will need to be developed. This could be an area of future research as well since there is a need to conduct field trials to see the practical application of this experiment, its further application to other medical protocols even outside that of the neonatal specialty, and its actual practical usefulness which further paves way for two potential directions for the evolution of cognitive AI -with applications in healthcare: monitoring and warning. It will be interesting to see if an AI coach can automatically recognise what is happening in the medical room through computer vision and recognise flaws in protocols, thus recognise if an action is performed or not, choose a pathway of treatment in its mental model which leads to the protocol being performed correctly and continue with the appropriate corresponding support signals to the health care practitioners.

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Declaration of interests

- ☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- ☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: